HAWEIA

Discover Saudi Arabia's rich heritage and embrace its future with Haweia. Explore breathtaking sights, book tickets, and unlock the treasures of this extraordinary land.

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Section: AAD

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Abstract

With the tourism sector evolving in the Kingdom and tourist indicators developing, along with the launch of pioneering projects such as the Boulevard, car races, Expo, etc. there is a noticeable interest in modern attractions. As citizens, we have observed a gap between tourists and their knowledge about the rich history of Saudi Arabia which poses a significant problem in order to achieve 2030 vision goals. Due to its vast geographical expanse, cultures, customs, architectural styles, and lifestyles vary. We propose the solution as consolidating everything related to cultural tourism in the Kingdom and its heritage, which could greatly alleviate this issue. Tourists from different parts of the world will not struggle to gather information from various sources, nor will they worry about its reliability. The solution will have a very positive impact on the historical perception of the Kingdom among different tourists, increase citizens income, and Saudi Arabia’s economy.

Introduction

Haweia is a unified platform of information developed to bridge the divide between tourists and the deep historical legacy of Saudi Arabia. Tourism in Saudi Arabia is now one of the important goals of the 2030 vision, and due to the diverse geographical regions, cultures, architectural styles, and ways of life, as well as the lack of apps that guide tourists to explore the Kingdom's rich tapestry, there is a need for a comprehensive solution. To address this problem, we have developed the Haweia system using the waterfall methodology. The outcome of this project are four fully implemented and tested functions that support our idea. Our system offers tourists the ability to enroll in volunteer opportunities, buy local products, and book events. Additionally, they have the choice to cancel their enrollment or save events to their favorites and book them later. This solution will no longer force travelers to struggle to find reliable information from scattered sources, book events online, explore different local products, or enroll in volunteer opportunities. Through Haweia, we will support citizens by increasing their income, support the 2030 vision goals, and provide a comprehensive platform for tourists to visit and plan their trip effectively.

Problem Statement

Many people are unaware of the Kingdom's domestic tourism opportunities and the rich heritage our country holds. This lack of awareness results in missed opportunities for individuals to explore and appreciate their own cultural heritage. Additionally, it poses challenges for the tourism sector in attracting both domestic and international visitors. Without a thorough understanding of the Kingdom's tourism potential and cultural significance, individuals miss out on enriching experiences, and the tourism industry faces limitations in its growth and contribution to the economy.

Proposed Solution

Our goal is to promote and highlight the rich culture and heritage of our Kingdom by providing an engaging and informative experience for users to explore. By creating an application, we aim to educate individuals about the diverse volunteer opportunities, products made by citizens and historical importance within the Kingdom. We seek to promote local tourism and foster a deep appreciation for our cultural heritage with immersive features and accessible information.

Approach

We utilized waterfall methodology to develop Haweia system. The waterfall consists of five phases, each phase has specific deliverables and must be completed before moving on to the next, ensuring clear progress tracking and control.

Below are the steps we followed in order to complete our project:

1-Requirements analysis and definition

After identifying the problem, we started to gather requirements of the project. We chose two ways, the interview and a survey. We interviewed the stakeholder to be able to gather important requirements for the system from the system users and their needs. Also, we created a questionnaire to collect quantitative data from large group of participants and those interested in the system. Then we created use case diagram to capture functional requirements and ensure that our system will deliver the desired functionalities and we have provided a description of each use case in the system. After that we draw a sequence for each use case to identify objects and the interaction between them.

2- System design

In this phase, we created the system design based on the requirement gathered in the pervious phase.We have created the architecture model and class diagram. For architecture we chose the repository pattern because our system depends on heavily data and the components are independent. Then we draw a class diagram with all attributes and methods to facilitate the process of selecting and defining classes and methods in the implementation phase.

3- Implementation

In this phase, based on the requirements phase and design phase we chose four functions to implemented in the system by using java language. The code was implemented by the following group members:

- **Rafal** created **Enrollmentopportunity()**
- **Ftoon** created **cancelEnrolment()** method & the **Customer class**
- **Lojain** created **Askforfavoriteevents()** method
- **Samia** created **Askforbuyproducts()** method

4- Testing

The system was tested by automated testing, to ensure that the functions of the system are working correctly and meet requirements quickly and efficiently. **All methods have been tasted by Shahad.**

Requirements Phase

We chose interview and questionnaire to gather project requirements because they offer distinct advantages. The interview provided detailed and qualitative insights through open discussions, allowing us to delve into nuances and gain a deeper understanding of stakeholders' perspectives. Additionally, the questionnaire enabled us to efficiently collect quantitative data from a wider range of participants, ensuring broader representation and facilitating statistical analysis. By using both methods, we aimed to ensure comprehensive coverage of the project requirements, leveraging the strengths of each approach for effective decision-making and project planning.

First technique: interview

Interviewee:	**Interviewers:**
Shahad Alharbi	Lojain Alghamdi Rafal Fakeera Ftoon Alsefri Samia Alhassni
Location\Medium:	**Appointement Date:**
Al-Safa, Jeddah 23455	Start Time: 13 March 5:30pm End Time: : 13 March 6:30pm
Objectives:	**Reminders:**
The objective of the interview is to gather crucial information from system users in order to determine the requirements for a specific system.	
General Observations:	
The interviewee responds, lean forward, maintain eye contact, and speak with a confident and assertive tone.	

Question 1:	Answer:
In general, do you think that cultural tourism in the Kingdom is the most attractive or not?	With the development of the Kingdom and the qualitative shift in tourism, we have noticed a great tendency for tourists abroad to go to places whose sites have spread on social media. Therefore, there is a deficiency regarding tourists' knowledge of the history of the Kingdom and its heritage landmarks.
Question 2:	**Answer:**
How do you think we can solve this problem?	By creating a website that encompasses all heritage sites, where visitors can view high-resolution images and detailed information about them (such as their story, age, purpose of creation, and more), in addition to mapping their locations so that visitors can understand the geographical expanse of the kingdom and choose the region they prefer according to their preferences.
Question 3:	**Answer:**
If you had the opportunity to make improvements to the website to make it more interactive for users, what would you add?	I would add a schedule containing all the events that may take place with information about them, this would help the tourist to schedule this trip easily. For the tourists to set a schedule for visiting the events they want. Creating a table like this helps show the diversity the Kingdom has. I would like to add more to this field and draw your attention to the fact that there are volunteer opportunities as well. I believe that tourists' participation in them will certainly enhance an emotional imprint in the tourist's memory.

Question 4:	Answer:
We honestly think this is a brilliant idea and would indeed contribute to addressing the problem of low tourism in heritage sites. However, we have a question; we want to make this proposed site an official destination for all tourists. What can you add to make this site the official destination for them in cultural tourism so that they do not need to visit other sites?	I would like to add the option of booking tickets as well instead of searching for ticket purchase locations yourself. This way, it would make it easier for tourists to engage in the activities they want without worry.
Question 5:	**Answer:**
What information do you prefer tourists to see about events and volunteer opportunities, whether they want to book or not? Also, regarding booking information, is presenting the price, time, and date sufficient?	I want you to present a description of the event/opportunities being held, including what will be offered, what activities people will engage in, mentioning if there are various shows such as musical performances, acrobatics, and others, the occasion for hosting this event, and some beautiful pictures of it. As for purchasing information, I also want to add an option for tourists to bookmark the event and be able to see the remaining tickets if the event is limited.
Question 6:	**Answer:**
Regarding volunteer opportunities, you previously mentioned that you want them in a schedule along with events. Do you want them only displayed with basic information, or do you want visitors to be able to see details and registration requirements and enroll online? Please specify all the details in this regard.	Yes, I want details about volunteering, registration requirements, time, and date to be displayed, and visitors should be able to register online because I believe this will make it easier for visitors to learn more about the kingdom and also reduce the hassle of searching, both for citizens and tourists.

Question 7:	Answer:
You mentioned at the beginning of the interview about volunteer opportunities and your interest in emotions and memories. What can we add to the website so that tourists can interact particularly with locals and take memories of the kingdom back home with them?	I believe that direct interaction between tourists and locals is essential for gaining insight into different cultures. Along with that, I also think it's important for tourists to take home souvenirs from their travels. To achieve this, I suggest adding a feature to the website that showcases local shops and their products, organized by categories for easy browsing. Each product would have a description, and information about the shop would also be included.
Question 8:	**Answer:**
Are there any additions you would like to make to the website to enhance the experience of tourists within their homes?	Yes, I would like to add virtual tours for events that tourists would like to see from their homes, where they can interact by zooming in and out, among other features. This would enhance the tourist experience and reflect the technological advancement of the kingdom, aligning with Vision 2030.

Second technique: survey

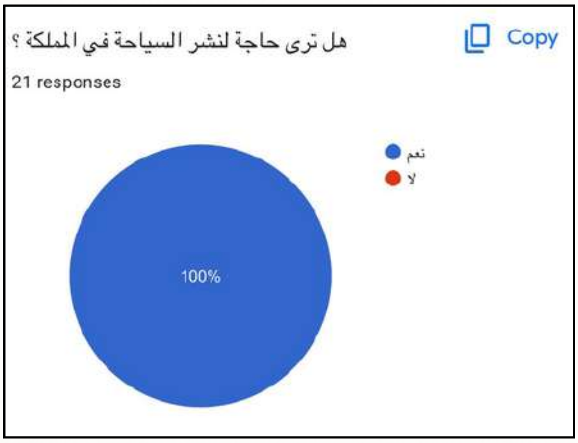


Figure 1: Survey results

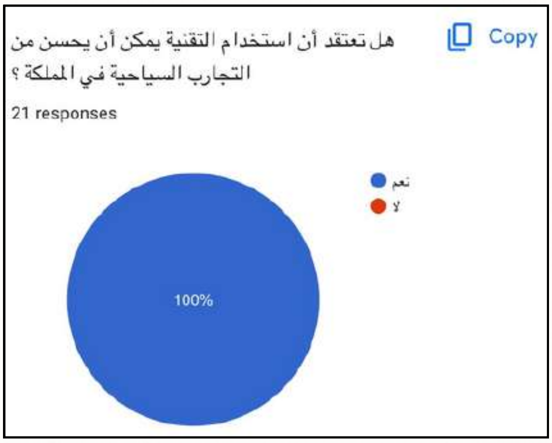


Figure 2: Survey results

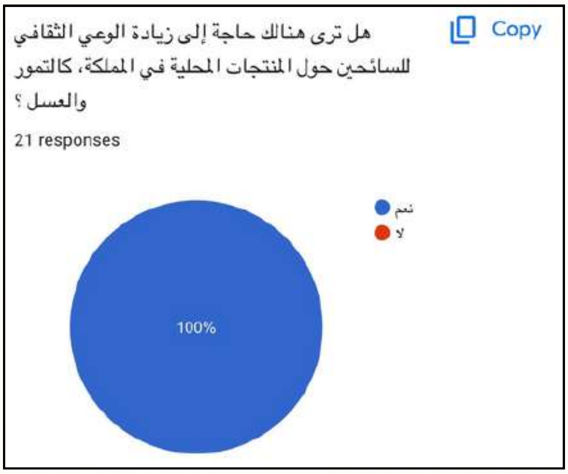


Figure 3: Survey results

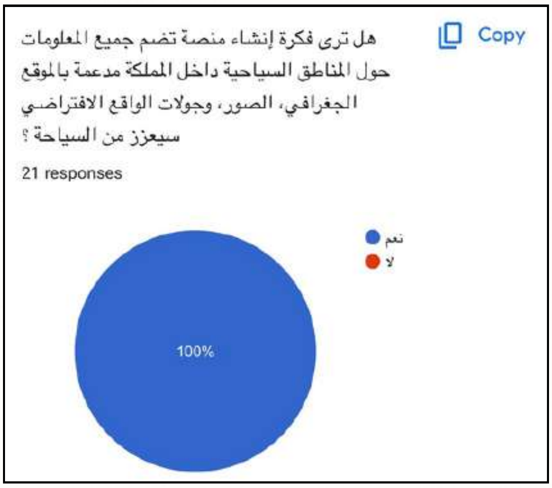


Figure 4: Survey results

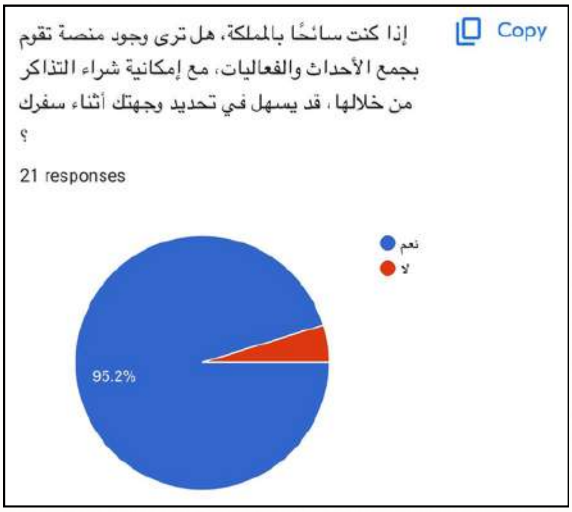


Figure 5: Survey results

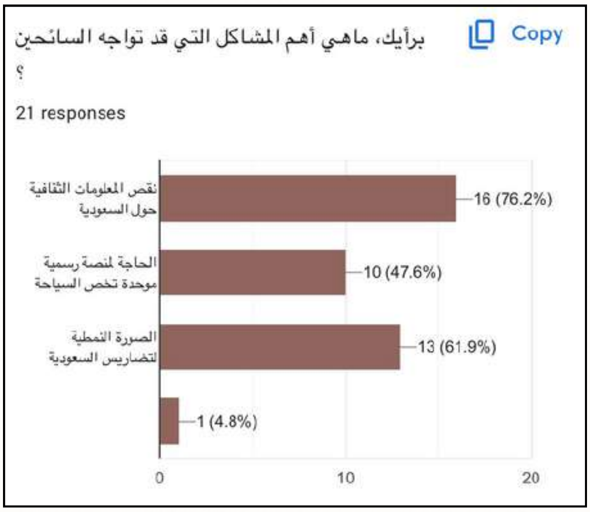


Figure 6: Survey results

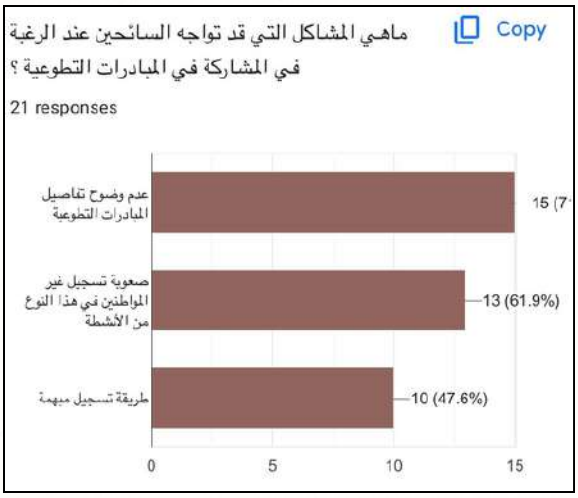


Figure 7: Survey results

Requirements Analysis

Based on the interview and survey outcomes, we have determined the following requirements:

1. The system shall display images of cultural heritage in the Kingdom.
   - 1.1 The system shall have a database of cultural heritage in the Kingdom.
   - 1.2 The system shall be capable of retrieving and displaying images of cultural heritage, historical, and cultural landmarks sites from the database.
   - 1.3 The system shall display information about the sites along with the images (e.g., site location, historical story, brief description)
   - 1.4 The system allows users to search for specific sites or browse through the images.
   - 1.5 The system allows users to select a specific city to view its sites.
2. The user can view all the events and volunteer opportunities in a calendar according to the selected month, and see the all details of the selected event.
   - 2.1 the system shall organize the collected events and opportunities in a calendar.
   - 2.2 the system shall present the events and opportunities to the user according to the month selected by the user.
   - 2.3 the system shall display all the details of the selected events by the user.
3. Enable users to book event tickets and receive notifications.
   - 3.1 When selecting a specific event, the system shall display comprehensive information about the event such as date, location, and ticket price, including a description and pictures.
   - 3.2 The system shall allow users to save events they wish to attend to their favorites list for viewing later.
   - 3.3 The system shall allow users to purchase tickets for the specific event.
   - 3.4 The system shall reserve a ticket once the date, time, and number of tickets are specified.
   - 3.5 The system shall regularly update events and opportunities.
   - 3.6 The system shall send notifications to users as soon as new updates occur.
4. The user can enroll in volunteer and community initiative opportunities related to heritage in the Kingdom.
   - 4.1 When a user selects a specific volunteer program, the system shall display the required qualifications, date, and time about the program and any additional requirements or restrictions.
   - 4.2 The system shall allow the user to enroll in the selected volunteer opportunity.
   - 4.3 The system shall display the information of the enrollment (e.g., volunteer's name, age, volunteer date).
   - 4.4 The system should allow users to submit their enrollment online.
   - 4.5 The system should have a notification mechanism to inform users about a new volunteer programs, and program updates.

5. Enable users to access virtual tours with 360-degree photos.
    5.1 The system allows the user to select a specific place to see the 360-degree photos.
    5.2 The system must integrate a viewer capable of displaying 360-degree photos seamlessly within the user interface.
    5.3 The system will allow the user with features such as panning, tilting, and zooming.
6. Customers should be able to purchase locally-made Saudi products.
    6.1 The system should display detailed product/ shop information, including product descriptions, images, and prices.
    6.2 The system should allow customers to add products to a shopping cart and proceed to checkout.
    6.3 The system should facilitate secure and convenient online payment options for customers to purchase the products.

Use case diagram

Use Case diagrams visually represent how users interact with a system to achieve specific goals. They clarify system functionality, identify requirements, facilitate communication, guide design, and support testing.

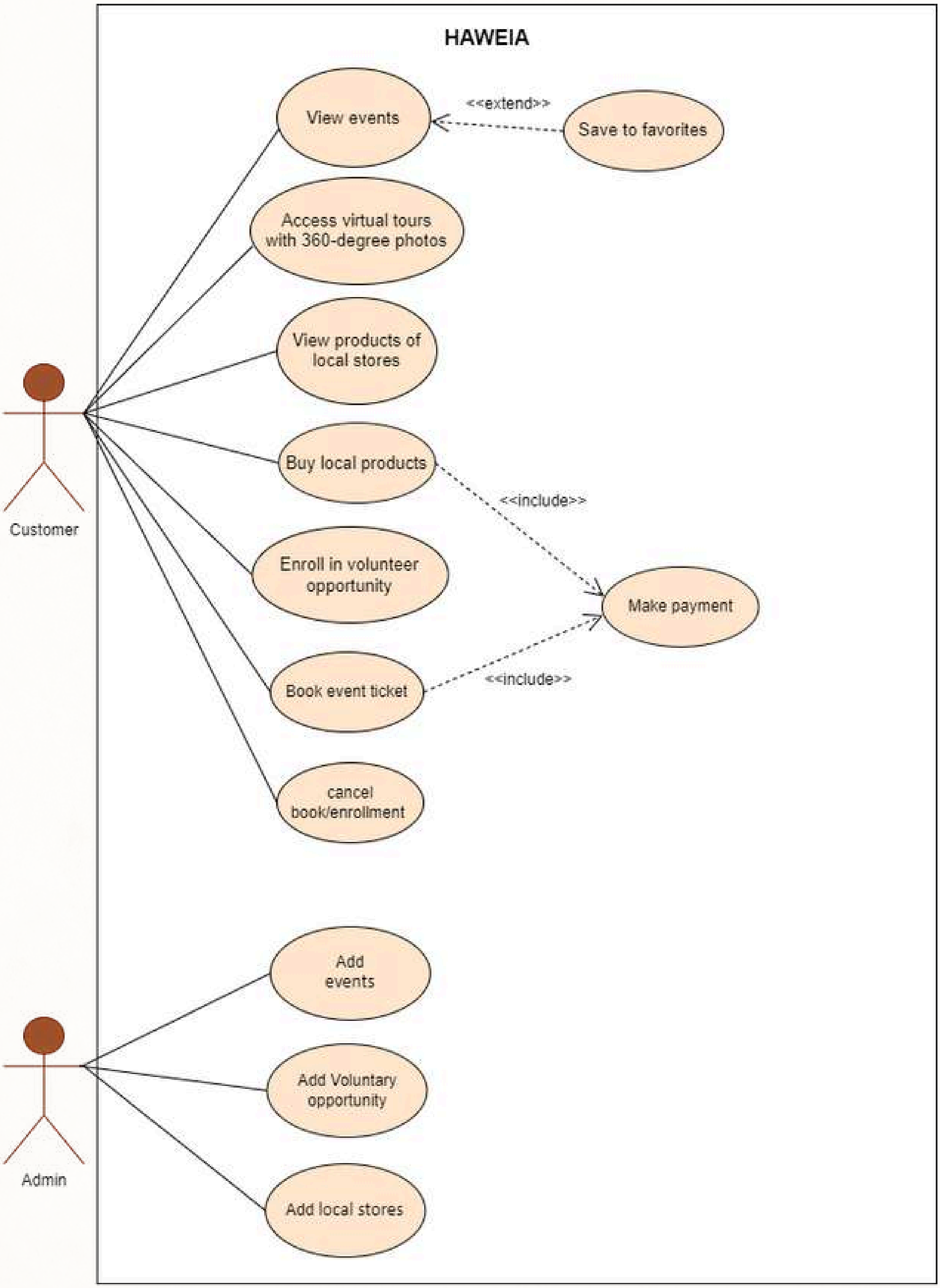


Figure 8: Use case diagram

Use Case Description:

Done by Rafal Fakeera

HAWEIA System: View events	
Actors	Customer
Description	The customer can access information about various events, explore upcoming events, obtain details such as dates, locations, descriptions, and save some to the favorite list.
Data	The chosen event
Stimulus	The user navigates to the events section to access detailed information.
Response	• The system displays a list of upcoming events.
Comments	

HAWEIA System: Cancel booking/ enrollment	
Actors	Customer
Description	The customer may decide to cancel his/ her booking or enrollment. cancellation may or may not be accepted according to his/ her number of canceled bookings/ enrollments for 3 months.
Data	The event to be canceled
Stimulus	Enter booked events section
Response	• The booking/ enrollment has been successfully canceled. • The cancellation has failed.
Comments	The user must confirm his/her cancellation when a confirmation message appears.

Use Case Description:

Done by Shahad Alharbi

HAWEIA System: Enroll in volunteers opportunities	
Actors	Customer
Description	The customer can view the volunteer opportunities and see the submission requirements, the opportunityname,ID, description, the period (start date - end date),capacity, and the volunteering location.
Data	The chosen volunteers opportunity
Stimulus	The user navigates to volunteers opportunities and choose one of them
Response	• a page of all opportunity details and requirments appear
Comments	the user must confirm submitting the requirements when confirmation message appears

HAWEIA System: Add local stores	
Actors	Admin
Description	The admin adds local stores, their products, and descriptions of each product
Data	(Id, name, and description
Stimulus	The administrator navigates to local stores and their products
Response	The system saves new inputs, validates, handles errors
Comments	The administrator must be logged in and have the permission to add local stores

Use Case Description:

Done by Lojain Alghamdi

HAWEIA System: View products of local stores	
Actors	Customer
Description	The customer will choose a specific products and then put it in the cart
Data	The chosen products
Stimulus	When customers open HAWEIA store
Response	• The Customer products will be added to the cart for the customer to pay
Comments	

HAWEIA System: : Add volunteers opportunities	
Actors	Admin
Description	This use case describes the process by which an administrator can add volunteers opportunities to the system. volunteers opportunities could include various activities or programs
Data	opportunities details (name, description, dates, objectives, audience, stakeholders, other info)
Stimulus	The administrator decides to add a new volunteers opportunities to the system
Response	• Upon submission, the system validates, saves, confirms, handles errors, recovers for connection loss
Comments	The administrator must be logged into the system and have the necessary permissions to add volunteers opportunities

Use Case Description:

Done by Samia Alhassni

HAWEIA System: Access virtual tours with 360-degree photos	
Actors	Customer
Description	Users have the ability to select a city of their choice and then proceed to explore virtual tours that showcase 360-degree images.
Data	City selection, virtual tour images.
Stimulus	Network availability and speed, and device capabilities.
Response	Display of Virtual Tours and images, loading 360-degree images, enabling user interaction.
Comments	

HAWEIA System: : Add Events	
Actors	Admin
Description	The admin has the ability to add new events and festivals to the system, providing details such as title, description, date, time, location, and images.
Data	Details of the event and festival.
Stimulus	The admin opens up events and festivals section.
Response	The system validates the input, stores the data, and confirms successful addition to the admin
Comments	

Use Case Description:

Done by Ftoon Alsefri

HAWEIA System: : Book event ticket	
Actors	Customer
Description	The user searches for the event he/she wants, and select the appropriate date and time, and more details about the event he/she selected, then he/she book the ticket.
Data	Event's details like time, and date, locations
Stimulus	The user select the book box
Response	Confirmation message for the book
Comments	

HAWEIA System: Make a payment	
Actors	Customer
Description	The user can making a payment process for a product from stores section or for an event ticket.
Data	Product/event details, user payment information(e.g., credit card details, billing address), payment amount.
Stimulus	User select product/event, view details then added to the cart, user enter payment information, user confirm the payment details and initiate the payment process.
Response	Payment analysed and status displayed (successful, pending, faile).
Comments	

Sequence Diagram

Sequence Diagrams visually depict the flow of messages between objects or components in a system over time. They clarify system behavior, logic, and collaboration patterns, aiding in design, implementation, testing, and debugging efforts.

Done by Rafal Fakeera

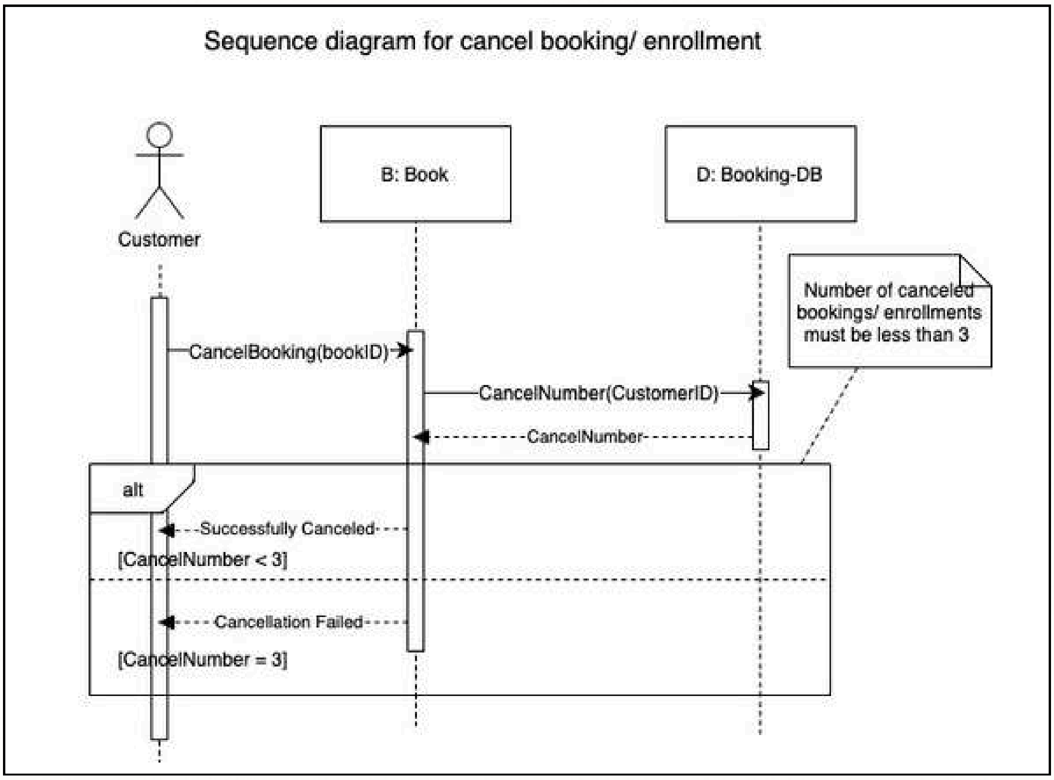


Figure 9: Sequence Diagram by Rafal

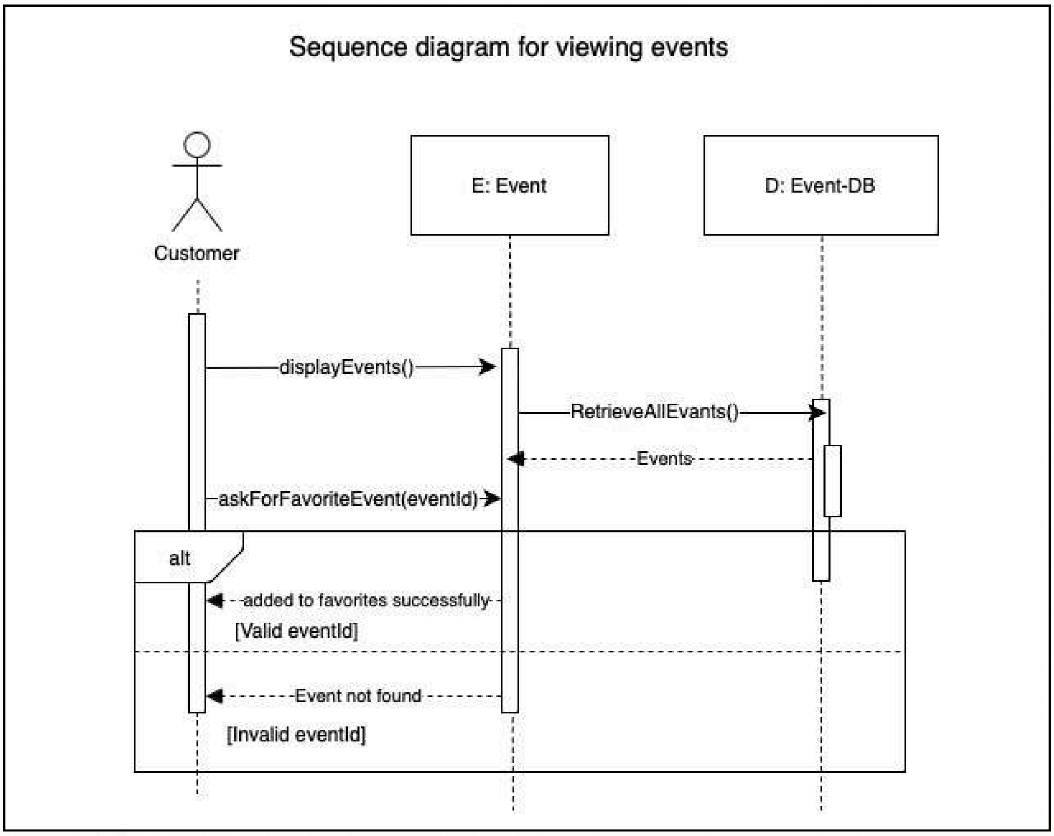


Figure 10: Sequence Diagram by Rafal

Sequence Diagram

Done by Lojain Alghamdi

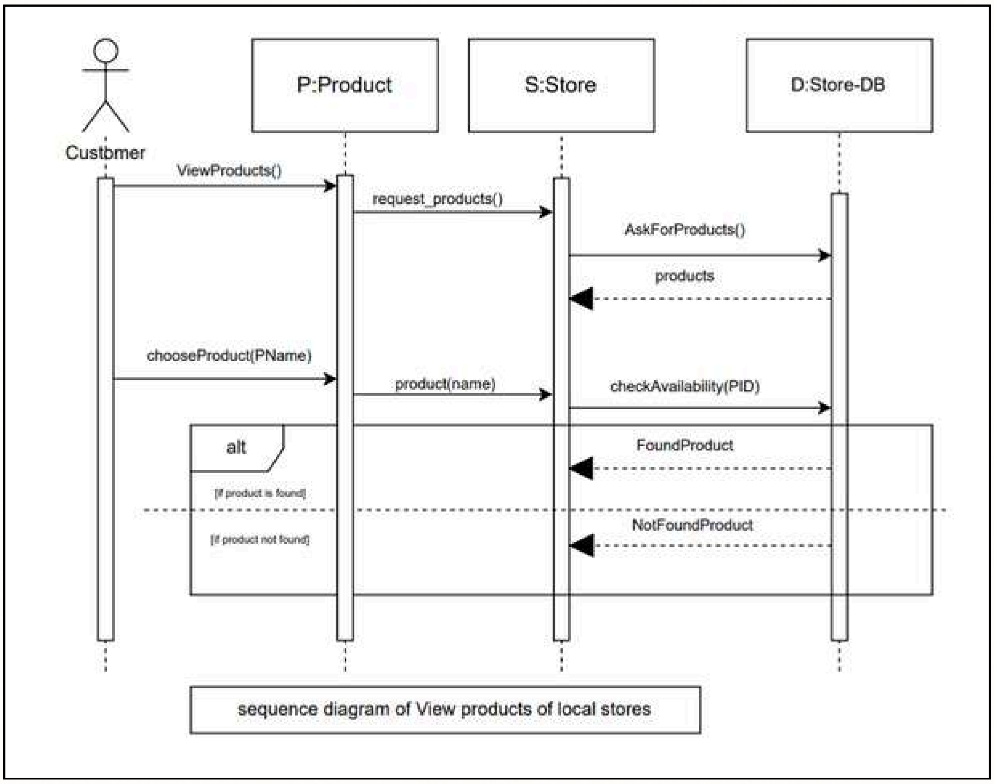


Figure 11: Sequence Diagram by Lojainl

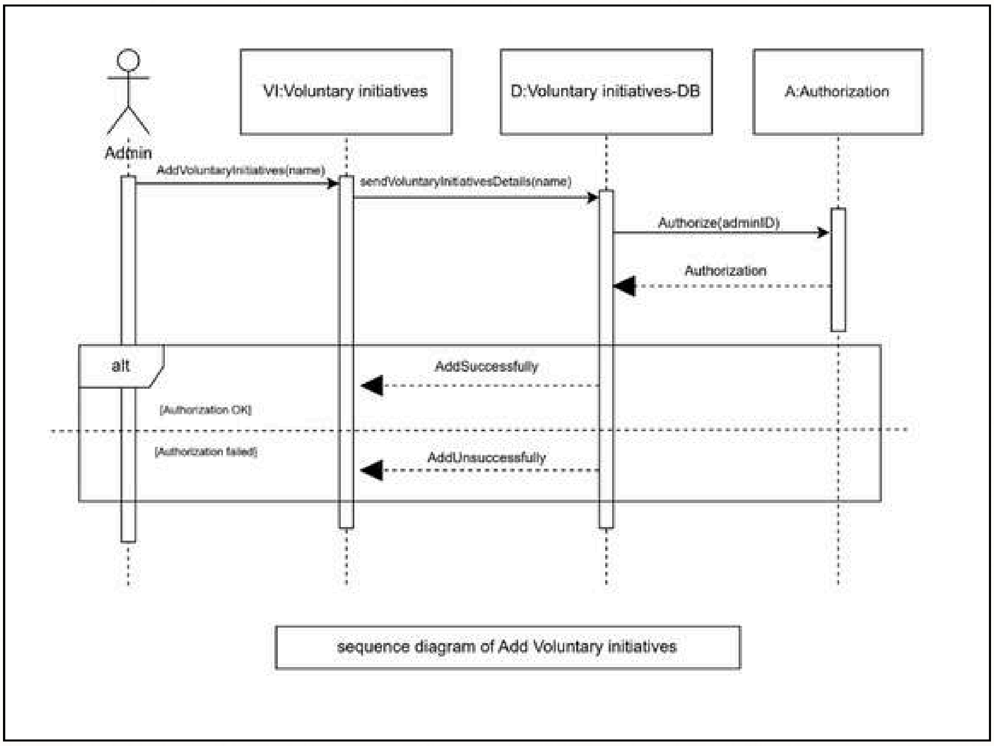


Figure 12: Sequence Diagram by Lojainl

Sequence Diagram

Done by Shahad Alharbi

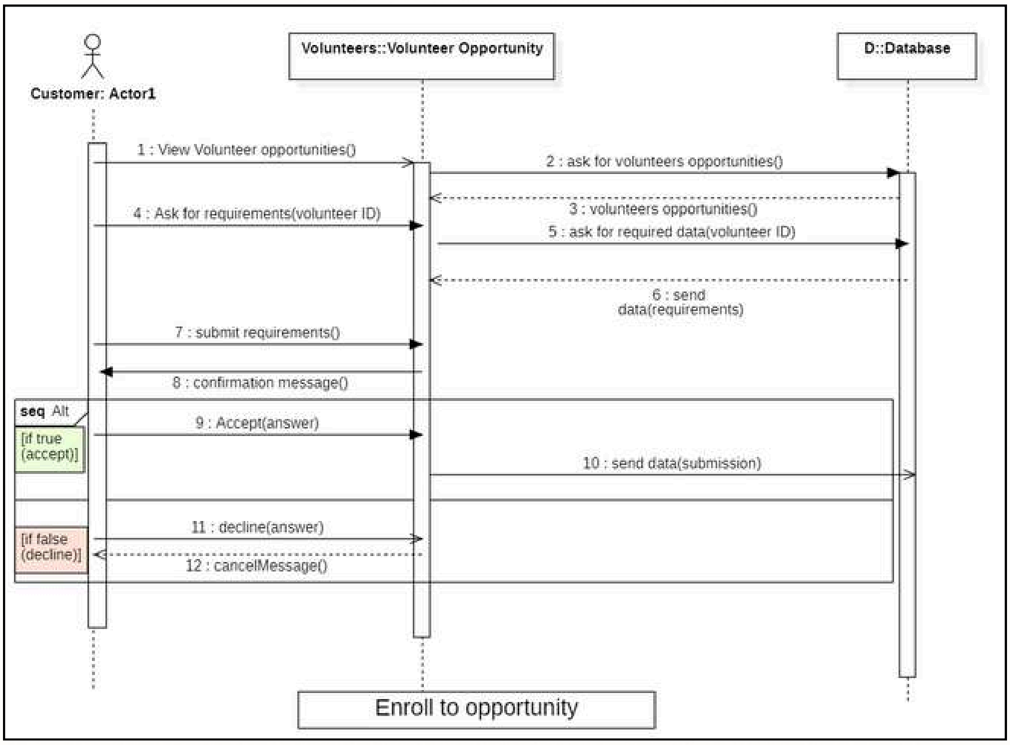


Figure 13: Sequence Diagram by Shahad

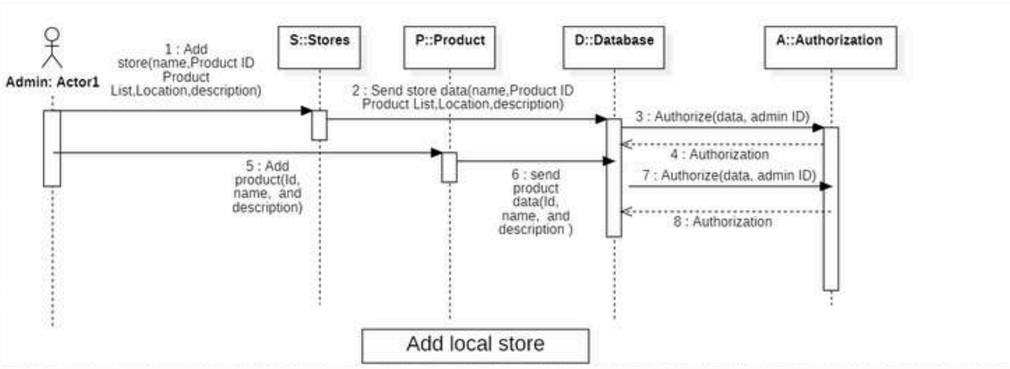


Figure 14: Sequence Diagram by Shahad

Sequence Diagram

Done by Ftoon Alsefri

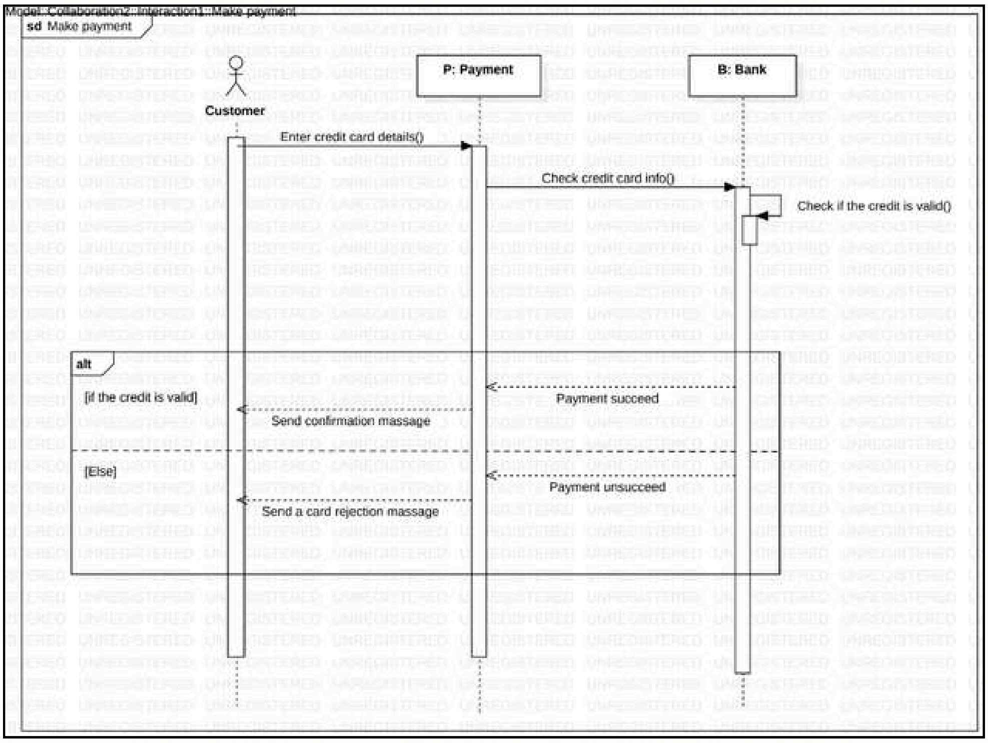


Figure 15: Sequence Diagram by Ftoon

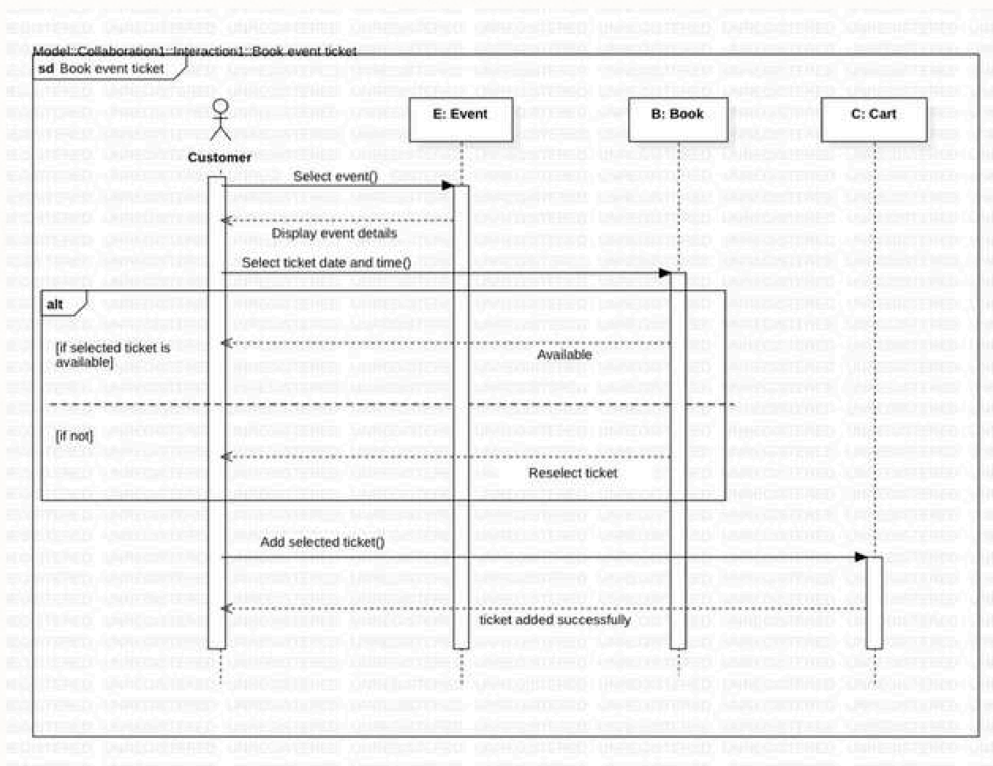


Figure 16: Sequence Diagram by Ftoon

Sequence Diagram

Done by Samia Alhassni

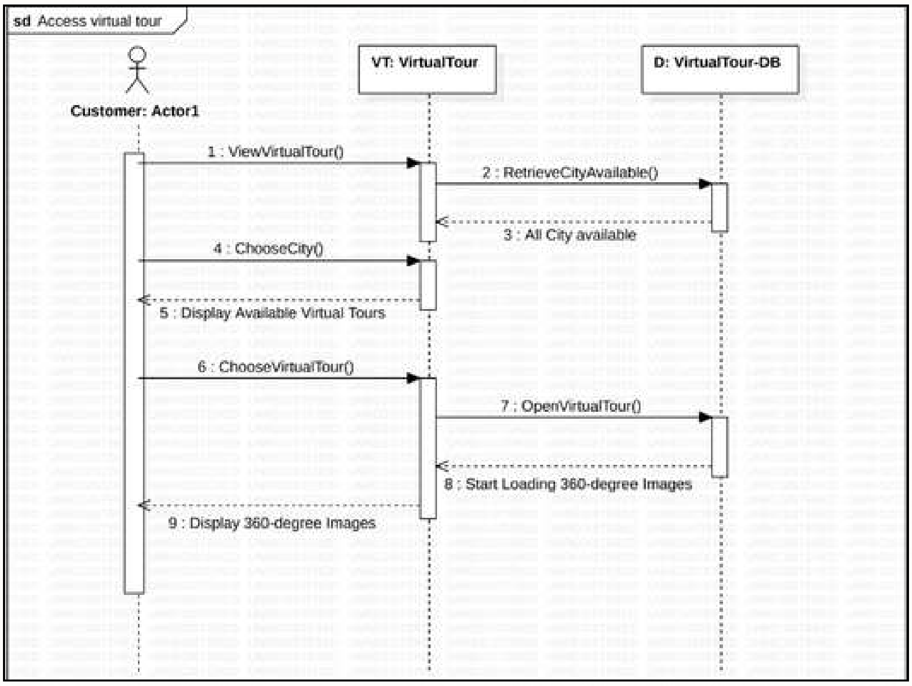


Figure 17: Sequence Diagram by Samia

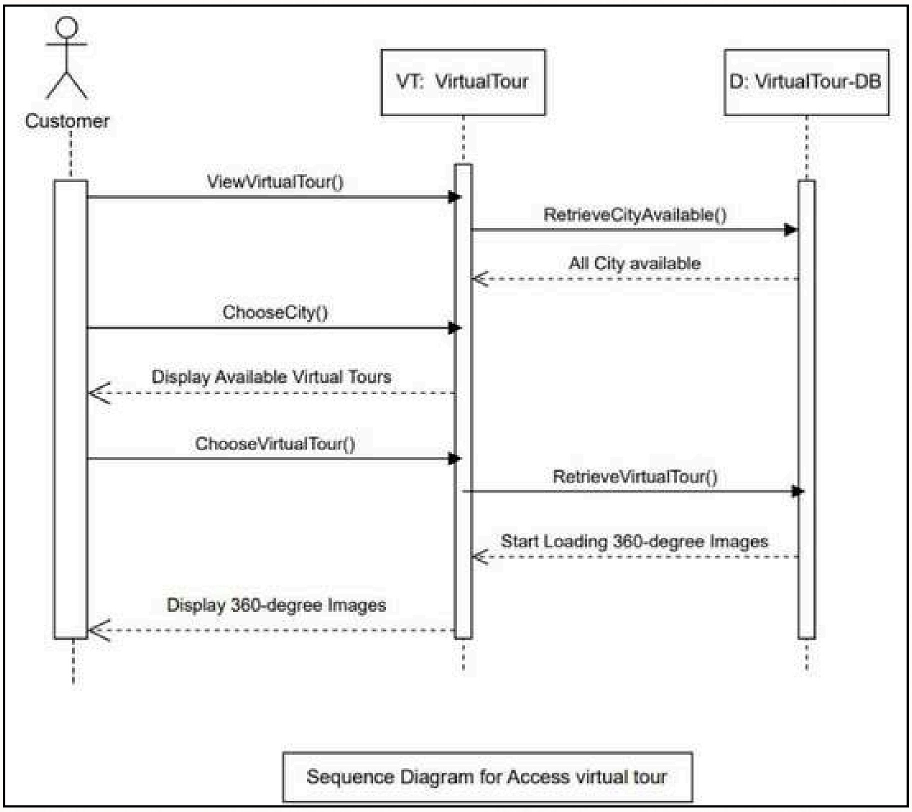


Figure 18: Sequence Diagram by Samia

Design Phase

Software architecture:

Software Architecture Diagrams visually depict the structure and components of a software system, aiding in understanding, communication, development guidance, decision-making, and documentation.

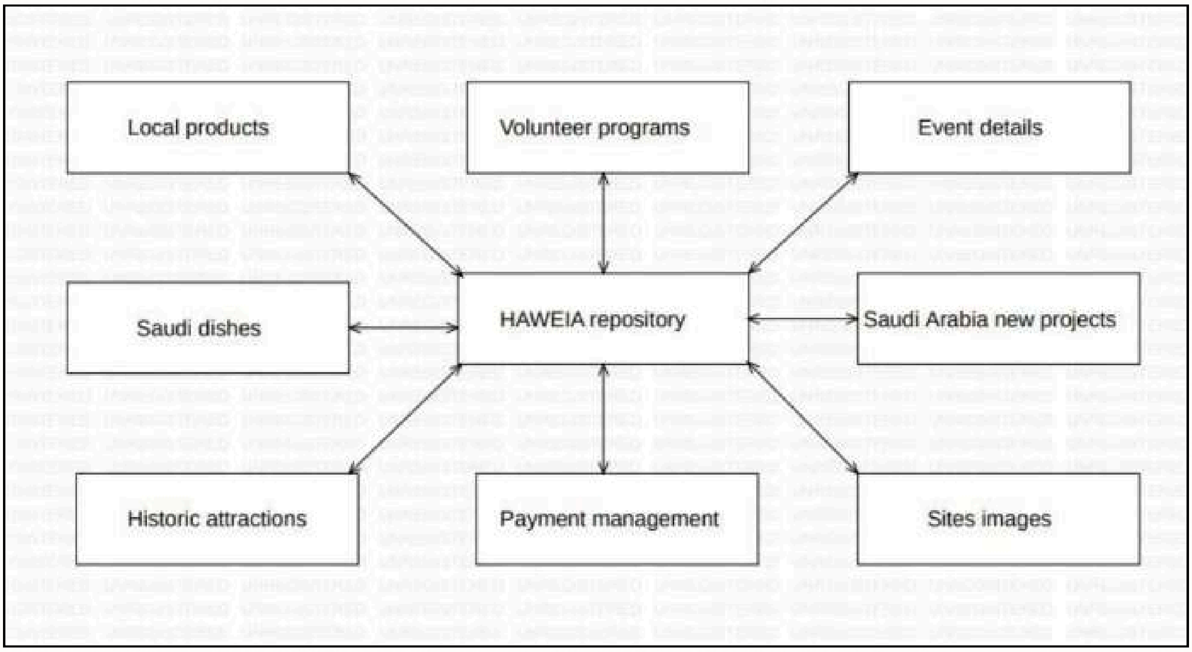


Figure 19: Software architecture

Our system relies heavily on data, such as images of cultural heritage, historical landmarks, and sites in the Kingdom, as well as a volunteer program including its objectives, activities, schedule, and any additional requirements or restrictions, etc. The Repository pattern is also:

- Used when a system has a large volume of information.
- Components can be independent.
- Data is managed consistently.

All these points are applicable to Haweia's system. That's why we believe that the Repository pattern is the most suitable pattern.

Design Phase

Class diagram:

Class Diagrams provide a visual representation of a system's structure, including classes, attributes, methods, and relationships. They clarify system components, relationships, and facilitate design and implementation decisions.

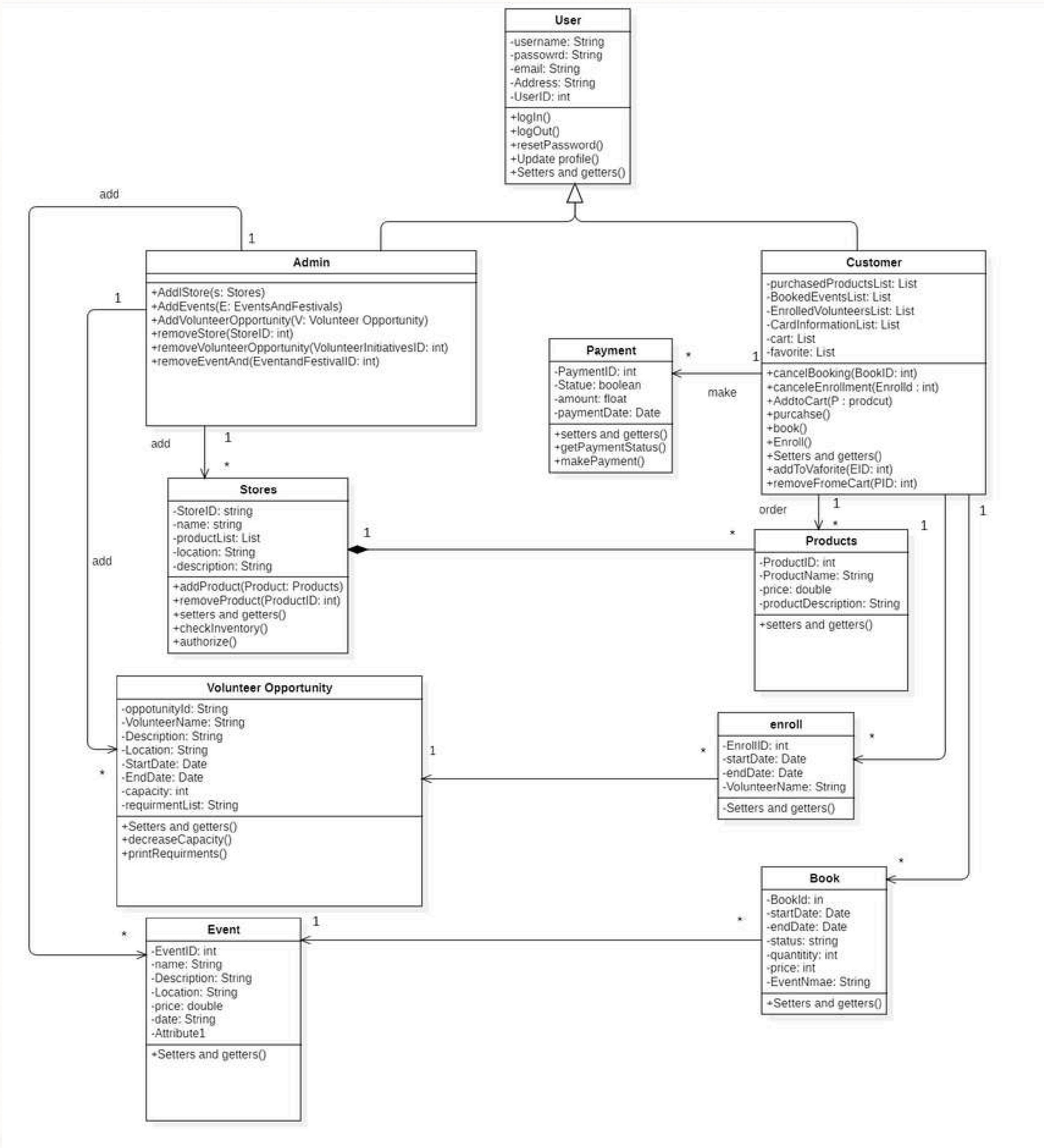


Figure 20: Class diagram

Implementation Phase

Our system consists of five classes: HaweiaProgram that have the main method, Event, Customer, Products, and Volunteer Opportunity

Classes:

- **HaweiaProgram class:** The main class contains several methods to manage and display events, volunteer opportunities, and products.
- **Event class:** Represents an event with details like name, date, location, description, and price.
- **Customer class:** Contains information such as the customer ID, Name, and list of favorite events, and a cart of products.
- **Products class:** Represents a product with details like name, price, and description.
- **VolunteerOpportunity class:** Represents a volunteer opportunity with details like description, location, requirements, capacity, and dates.

In the beginning, "displayWelcomeMessage()" method is called to print a welcome message. Then, volunteer opportunities, events, and products are initialized by calling the appropriate methods.

Inside the loop:

- promptUserForServiceChoice(): method is called to display a list of available services and allow the user to choose a service.
- Based on the user's choice, the "processServiceChoice(choice)" method is called to display details of the chosen service.

- If the chosen service is products, the "askForBuyProduct(productId)" method is called.
- If the chosen service is events, the "askForFavoriteEvent(eventId)" method is called.
- If the chosen service is volunteer opportunities, the "askForEnrollment()" method is called. If the user agrees, the opportunity ID is requested from the user, the requirements are checked using "isRequirementsFulfilled(opportunityID)", and then the user is enrolled via "enrollOpportunity(opportunityID, requirementsFulfilled)".

- After executing the service, the user is asked if they need another service. If the answer is "no" the loop breaks, and the program ends.

Implementation Phase

```java
public static boolean askForBuyProduct(int productId) {
    // Find the Product corresponding to the given Product ID
    Products ProductToBuy = null;
    for (Products product : products) {
        if (product.getproductId() == productId) {
          ProductToBuy = product;
          break;
        }
    }
    // Check if the Product is found
       if (ProductToBuy != null) {
        // Call the addToCart method of the Customer class with the product object
        Ahmed.addToCart(ProductToBuy);
        return true; // Product us found and added to cart

      }else {
        return false; // product not found
  }
}
```

Figure 21: askForBuyProduct(int productId) method

1- askForBuyProduct(int productId):
This method takes a productId as a parameter. It searches for the product with the given productId in the products list. If the product is found, it adds the product to the customer's cart using the addToCart method and returns true. Otherwise if the product is not found, it returns false.

```java
//Displays the cart products for a given customer.
public static void displayCart(Customer customer, Scanner scanner) {
    System.out.print("Do you want to buy the products? (y/n): ");
    String res = scanner.next();

if (res.equalsIgnoreCase("y")) {
    List<Products> cart = customer.getCart();

             if (cart.isEmpty()) {
                System.out.println("You have no product in your cart.");
              } else
                cart.clear();
                System.out.println("Cart items purchased successfully.");

    }
}
```

Figure 22: displayCart(Customer customer, Scanner scanner) methode

2- displayCart(Customer customer, Scanner scanner):
This method asks the customer if they want to buy the products in their cart. If the customer confirms with "y", it clears the cart and prints "Cart items purchased successfully." to confirm the purchase to the user. Otherwise, it prints "You have no product in your cart".

Implementation Phase

```java
public static boolean askForFavoriteEvent(int eventId) {
        // Find the event corresponding to the given Event ID
        Event eventToAdd = null;
        for (Event event : events) {
            if (event.getId() == eventId) {
                eventToAdd = event;
                break;
            }
        }

        // Check if the event is found
        if (eventToAdd != null) {
            // Call the addToFavorite method of the Customer class with the Event object
            Ahmed.addToFavorite(eventToAdd);
            return true; // Event found and added to favorites
        } else {
            return false; // Event not found
        }
    }
```

Figure 23:askForFavoriteEvent(int eventId) method

3- askForFavoriteEvent(int eventId)

This method searches for an event by its ID in a list of events. If found, it adds it to the customer's list of favorite events and returns true; otherwise, it returns false. The Event class represents an event entity with various properties like ID, name, date, etc.

```java
    //Method to cancel an opportunity:
    public static boolean cancelOpportunity(String opportunityId, Customer customer) {
        // Find the opportunity in the enrolled volunteer opportunities
        VolunteerOpportunity opportunityToCancel = null;
        for (VolunteerOpportunity opportunity : customer.getEnrolledVolunteer()) {
            if (opportunity.getOpportunityId().equalsIgnoreCase(opportunityId)) {
                opportunityToCancel = opportunity;
                break;
            }
        }

        if (opportunityToCancel != null) {
            // Check the number of cancellations
            if (customer.getCancelNumber() >= 3) {
                System.out.println("You cannot cancel the enrollment as you have reached the maximum number of cancellations.");
                return false;
            } else {
                // Remove the opportunity from the enrolled list
                customer.cancelEnrollment(opportunityToCancel);
                // Increase the cancellation counter
                customer.setCancelNumber(customer.getCancelNumber() + 1);
                // Increase the capacity of the opportunity after cancel
                opportunityToCancel.setCapacity(opportunityToCancel.getCapacity() + 1);
                System.out.println("Enrollment canceled successfully.");
                return true;
            }
        } else {
            System.out.println("Opportunity not found.");
            return false;
        }
```

Figure 24: cancelOpportunity(string opportunityId, Customer cutomer) method

4- cancelOpportunity(string opportunityId, Customer cutomer)

This method first will check if the id of the opportunity is found, it will check if the number of cancelation the customer do is => 3 it will return false and the cancel will not completed, if it is less than 3 it will cancel the opportunity and then increase the number of cancelation and the capacity of the opportunity after cancelation and return true. If the id of the opportunity is not found it will return false and the cancel will not completed.

Implementation Phase

```java
//This method will check if the requirements are fulfilled using isRequirementsFulfilled() method and the capacity using !isCapacityFull()
//If both are true, then enroll
public static boolean enrollOpportunity(String opportunityId, boolean requirementsFulfilled) {
    // Find the selected opportunity in the linked list
    VolunteerOpportunity selectedOpportunity = null;

    //Finding opportunity using opportunityId
    for (VolunteerOpportunity opportunity : Opportunity) {
        if (opportunity.getOpportunityId().equalsIgnoreCase(opportunityId)) {
            selectedOpportunity = opportunity;
            break;
        }
    }

    if (selectedOpportunity != null) {

        if (requirementsFulfilled) {

            if (!selectedOpportunity.isCapacityFull()) {
                Ahmed.enroll(selectedOpportunity);
                selectedOpportunity.decreaseCapacity();
                System.out.println("You have successfully enrolled in the volunteer opportunity.");
                return true;
            } else { //If capacity is full
                System.out.println("You cannot enroll in this opportunity because the capacity is full.");
            }

        } else { //If not all requirements are met
            System.out.print("You cannot enroll in this opportunity as not all requirements are met.");
        }

    } else { //If selectedOpportunity = null
        System.out.println("Invalid opportunity ID.");

    }

    return false;
}
```

Figure 25: enrollOpportunity(String opportunityId, boolean requirementsFulfilled)

5- enrollOpportunity(String opportunityId, boolean requirementsFulfilled)

This method takes two parameters, opportunityId as a String representing the unique identifier of the volunteer opportunity and requirementsFulfilled a boolean value indicating whether the user has fulfilled the requirements for the opportunity. The method first initializes a VolunteerOpportunity object and assigns it a null value. It then uses the opportunityId to find the corresponding volunteer opportunity, and checks if the opportunityId is valid and if the capacity of the selected opportunity is not full. If these conditions are met and the user has fulfilled the requirements, the user is successfully enrolled in the volunteer opportunity; otherwise, an appropriate error message is displayed.

```java
//Method to print each requirement and ask the user if he fulfill the requirement
public static boolean isRequirementsFulfilled(String opportunityId, String response) {

    VolunteerOpportunity selectedOpportunity = null;

    //Finding opportunity using opportunityId
    for (VolunteerOpportunity opportunity : Opportunity) {
        if (opportunity.getOpportunityId().equalsIgnoreCase(opportunityId)) {
            selectedOpportunity = opportunity;
        }
    }

    if (selectedOpportunity != null && response.equalsIgnoreCase("y")) {
        return true;
    }

    return false; // If requirements are not fulfilled, return false
}
```

Figure 26: isRequirementsFulfilled(String opportunityId, String response) method

6- isRequirementsFulfilled(String opportunityId, String response)

isRequirementsFulfilled method simply check if the customer fulfill the opportunity requirements by evaluating his answer. If the customer entered “y” then true is returned, otherwise false will be returned.

Implementation Phase

Screenshot of running code on the console window:

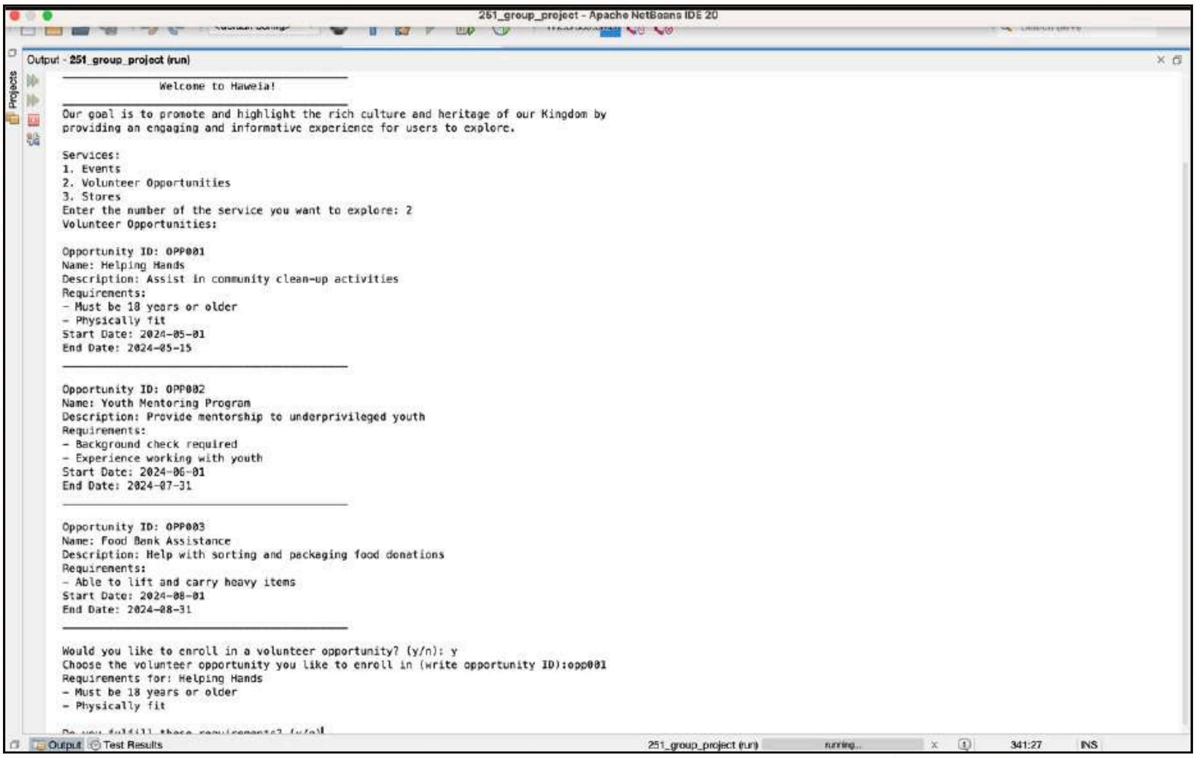


Figure 27: Running code on the console window

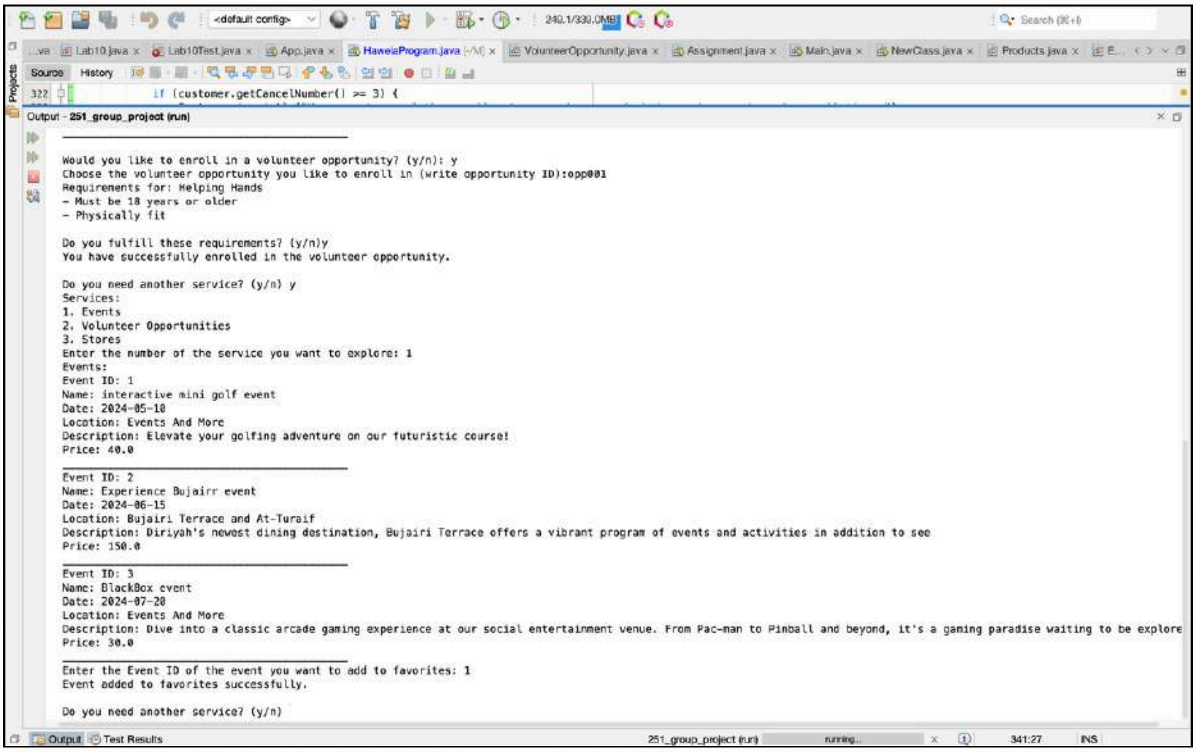


Figure 28: Running code on the console window

Testing Phase

During the test phase, we implemented seven methods. We used assertTrue/assertFalse to validate the tests. Additionally, we set up and added objects to the four classes to run the code.

Setup

first, we did the setup to test the rest of the methods and we made 4 objects from those classes (event, product, volunteer opportunity, Customer)

```
@Before
public void setUp() {
    Event event1 = new Event( id: 1,  name: "interactive mini golf event",  date: "2024-05-10",  location: "Events And More",  description: "Elevate y
    events.add( e: event1);

    Products product1 = new Products( productId: 1,  productName: "Qassim dates",  productPrice: 40,  productDescription: "A distinctive agricultural prod
    products.add( e: product1);

    LinkedList<String> requirements_opportunity3 = new LinkedList<>();
    requirements_opportunity3.add( e: "Able to lift and carry heavy items");
    VolunteerOpportunity opportunity3 = new VolunteerOpportunity(
            opportunityId: "OPP003",
            name: "Food Bank Assistance",
            description: "Help with sorting and packaging food donations",
            location: "Jeddah",
            capacity: 15,
            requirements: requirements_opportunity3,
            startDate: "2024-08-01",
            endDate: "2024-08-31"
    );
    Opportunity.add( e: opportunity3);


    Ahmed.enroll( volunteer: opportunity3);
}
```

Figure 29: Setup

AddToFavoriteEvent

we did two test methods, the first one will return true if the event ID is found, and the second one will return false because the event ID is not found, we tried to do more testing here like sending an invalid ID but we couldn't because the ID we are sending as a parameter and it's a syntax error if we tried

```
    /**
     * Test of displayWelcomeMessage method, of class HaweiaProgram.
     */

    @Test
    public void testaddToFavoriteEvent_IfIdFound() {
         System.out.println( x: "askForFavoriteEvent");
        int Id = 1;
        boolean result = HaweiaProgram.addToFavoriteEvent( eventId: Id);
        assertTrue(  condition: result);
        // TODO review the generated test code and remove the default call to fail.

    }
     @Test
    public void testaddToFavoriteEvent_IfIdNotFound() {
        System.out.println( x: "askForFavoriteEvent");
        int eventId = 5;
        boolean result = HaweiaProgram.addToFavoriteEvent(eventId);
        assertFalse(  condition: result);
        // TODO review the generated test code and remove the default call to fail.

    }
```

Figure 30: Testing AddToFavoriteEvent

Testing Phase

AddToCart

we did two test methods, the first one will return true if the product ID is found, and the second one will return false because the product ID is not found, we tried to do more testing here like sending an invalid ID but we couldn't because the ID we are sending as a parameter and it's a syntax error if we tried

```
@Test
public void testaddToCart_IfFaound() {
    int ProductId =1;
    boolean result = HaweiaProgram.addToCart( productId: ProductId);
    assertTrue( condition: result);
}

 @Test
public void testaddToCart_IfNotFaound() {
    int ProductId =7;
    boolean result = HaweiaProgram.addToCart( productId: ProductId);
    assertFalse( condition: result);
}
```

Figure 31: Testing AddToCart

EnrollOpportunity

we did three test methods, the first one will return true if the opportunity ID is found and requirements fulfilled, the second one will return false because the opportunity ID is found but the requirement isn't fulfilled, and the third one will return false because the ID is invalid regarding the requirements fulfillment

```
 * Test of enrollOpportunity method, of class HaweiaProgram.
 */
@Test
public void testEnrollOpportunity_IfIdFound_FulFilld() {
    String opportunityId = "OPP003";
    boolean result= HaweiaProgram.enrollOpportunity(opportunityId, requirementsFulfilled: true);
    assertTrue( condition: result);
}
@Test
public void testEnrollOpportunity_IfIdFound_NotFulFilld() {
    String opportunityId = "OPP003";
    boolean result= HaweiaProgram.enrollOpportunity(opportunityId, requirementsFulfilled: false);
    assertFalse( condition: result);
}
@Test
public void testEnrollOpportunity_IfIdNotValid() {
    String opportunityId = "OPPttyrey001";
    boolean result= HaweiaProgram.enrollOpportunity(opportunityId, requirementsFulfilled: true);
    assertFalse( condition: result);
}
```

Figure 32: Testing EnrollOpportunity

Testing Phase

CancelOpportunity

we did two test methods, the first one will return true if the opportunity ID is found and the opportunity is on the enrollments page of the customer, and the second one will return false because the opportunity ID is not found and the opportunity is on the enrollments page of the customer

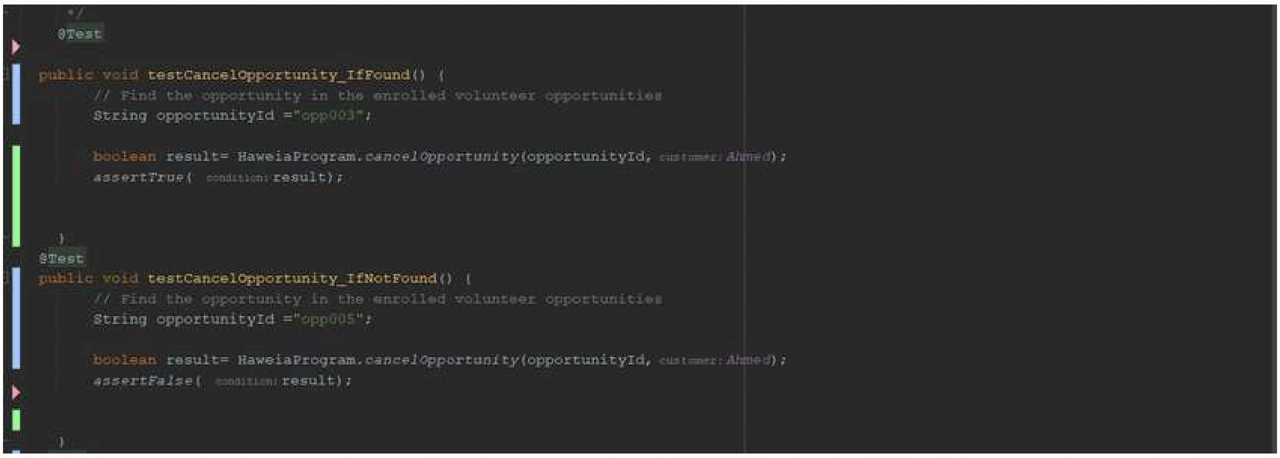


Figure 33: Testing CancelOpportunity

this is the test result after running the test file. All 9 test methods passed. On the right side, there are print messages inside some methods.

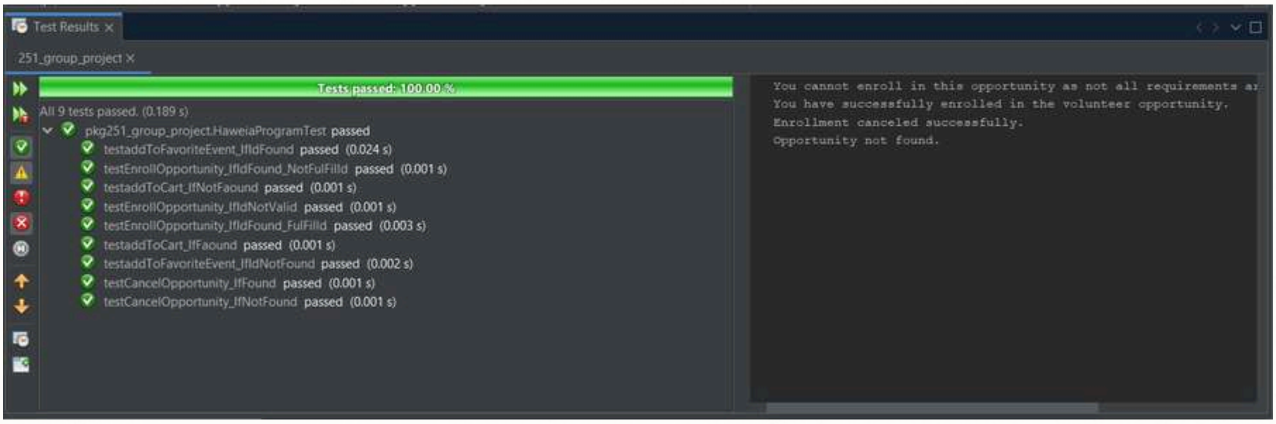
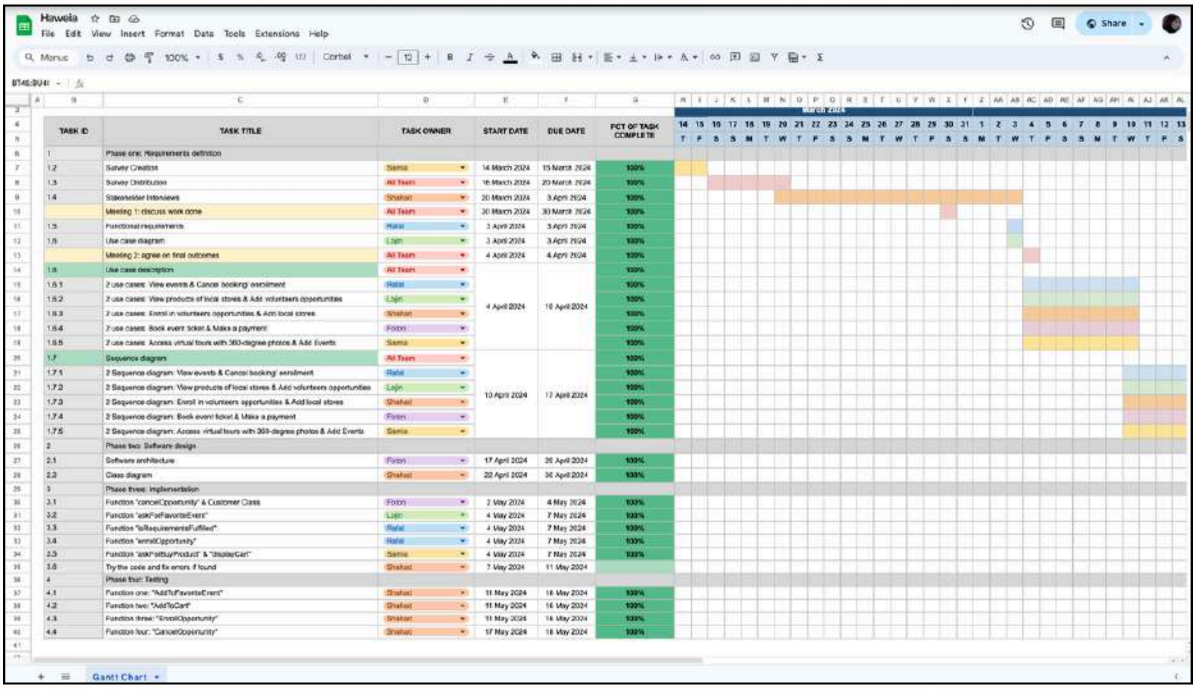


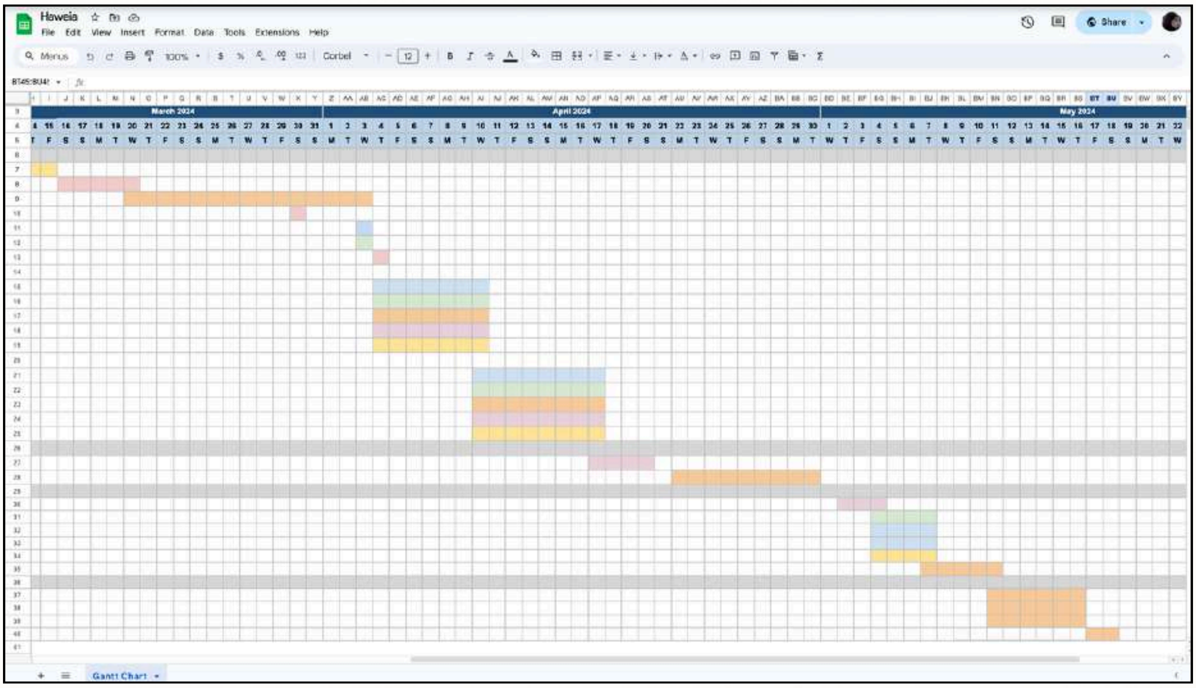
Figure 34: Testing code

Gantt Chart

We have created a Gantt chart to plan the devolopment process of Haweia system using Google sheet. Click here to view the entire sheet.



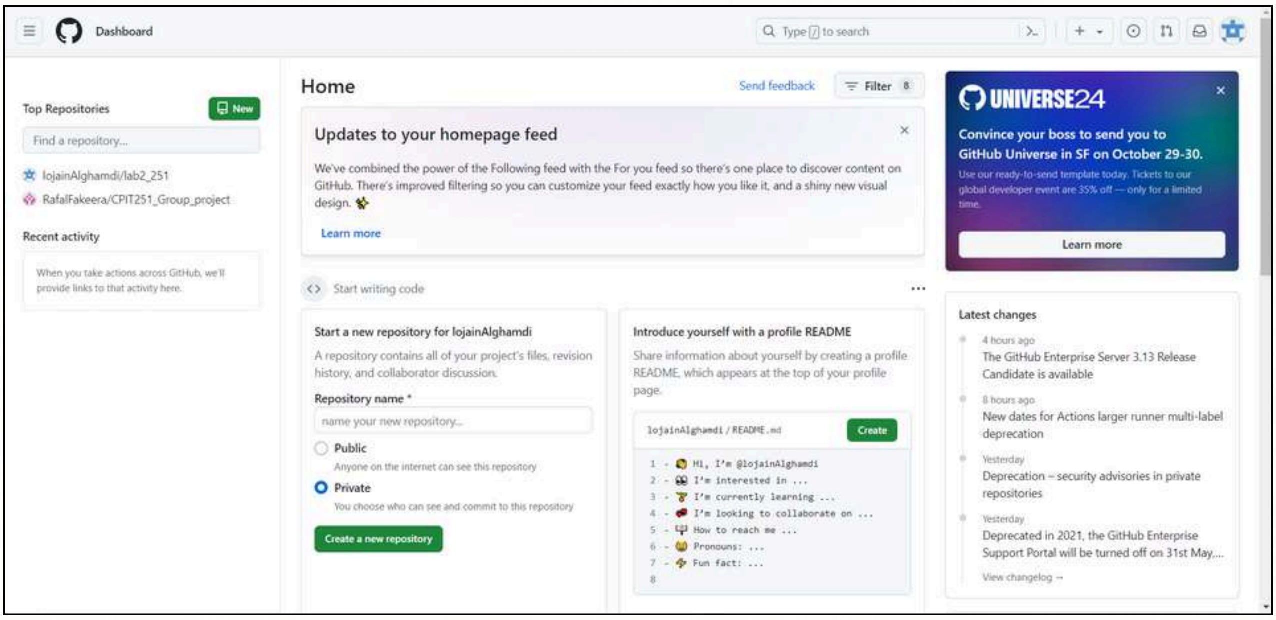
TASK ID	TASK TITLE	TASK OWNER	START DATE	DUE DATE	PCT OF TASK COMPLETE
1	Phase one: Requirements definition				
1.2	Survey Creation	Sama	14 March 2024	15 March 2024	100%
1.3	Survey Distribution	All Team	16 March 2024	20 March 2024	100%
1.4	Stakeholder Interviews	Shahad	20 March 2024	3 April 2024	100%
	Meeting 1: discuss work done	All Team	30 March 2024	30 March 2024	100%
1.5	Functional requirements	Rafal	3 April 2024	3 April 2024	100%
1.6	Use case diagram	Lajin	3 April 2024	3 April 2024	100%
	Meeting 2: agree on final outcomes	All Team	4 April 2024	4 April 2024	100%
1.6	Use case description	All Team	4 April 2024	10 April 2024	100%
1.6.1	2 use cases: View events & Cancel booking/ enrollment	Rafal			100%
1.6.2	2 use cases: View products of local stores & Add volunteers opportunities	Lajin			100%
1.6.3	2 use cases: Enroll in volunteers opportunities & Add local stores	Shahad			100%
1.6.4	2 use cases: Book event ticket & Make a payment	Foton			100%
1.6.5	2 use cases: Access virtual tours with 360-degree photos & Add Events	Sama			100%
1.7	Sequence diagram	All Team	10 April 2024	17 April 2024	100%
1.7.1	2 Sequence diagram: View events & Cancel booking/ enrollment	Rafal			100%
1.7.2	2 Sequence diagram: View products of local stores & Add volunteers opportunities	Lajin			100%
1.7.3	2 Sequence diagram: Enroll in volunteers opportunities & Add local stores	Shahad			100%
1.7.4	2 Sequence diagram: Book event ticket & Make a payment	Foton			100%
1.7.5	2 Sequence diagram: Access virtual tours with 360-degree photos & Add Events	Sama			100%
2	Phase two: Software design				
2.1	Software architecture	Foton	17 April 2024	20 April 2024	100%
2.2	Class diagram	Shahad	22 April 2024	30 April 2024	100%
3	Phase three: Implementation				
3.1	Function "cancelOpportunity" & Customer Class	Foton	2 May 2024	4 May 2024	100%
3.2	Function "askForFavoriteEvent"	Lajin	4 May 2024	7 May 2024	100%
3.3	Function "isRequirementsFulfilled"	Rafal	4 May 2024	7 May 2024	100%
3.4	Function "enrollOpportunity"	Rafal	4 May 2024	7 May 2024	100%
3.5	Function "askForBuyProduct" & "displayCart"	Sama	4 May 2024	7 May 2024	100%
3.6	Try the code and fix errors if found	Shahad	7 May 2024	11 May 2024	
4	Phase four: Testing				
4.1	Function one: "AddToFavoriteEvent"	Shahad	11 May 2024	16 May 2024	100%
4.2	Function two: "AddToCart"	Shahad	11 May 2024	16 May 2024	100%
4.3	Function three: "EnrollOpportunity"	Shahad	11 May 2024	16 May 2024	100%
4.4	Function four: "CancelOpportunity"	Shahad	17 May 2024	18 May 2024	100%



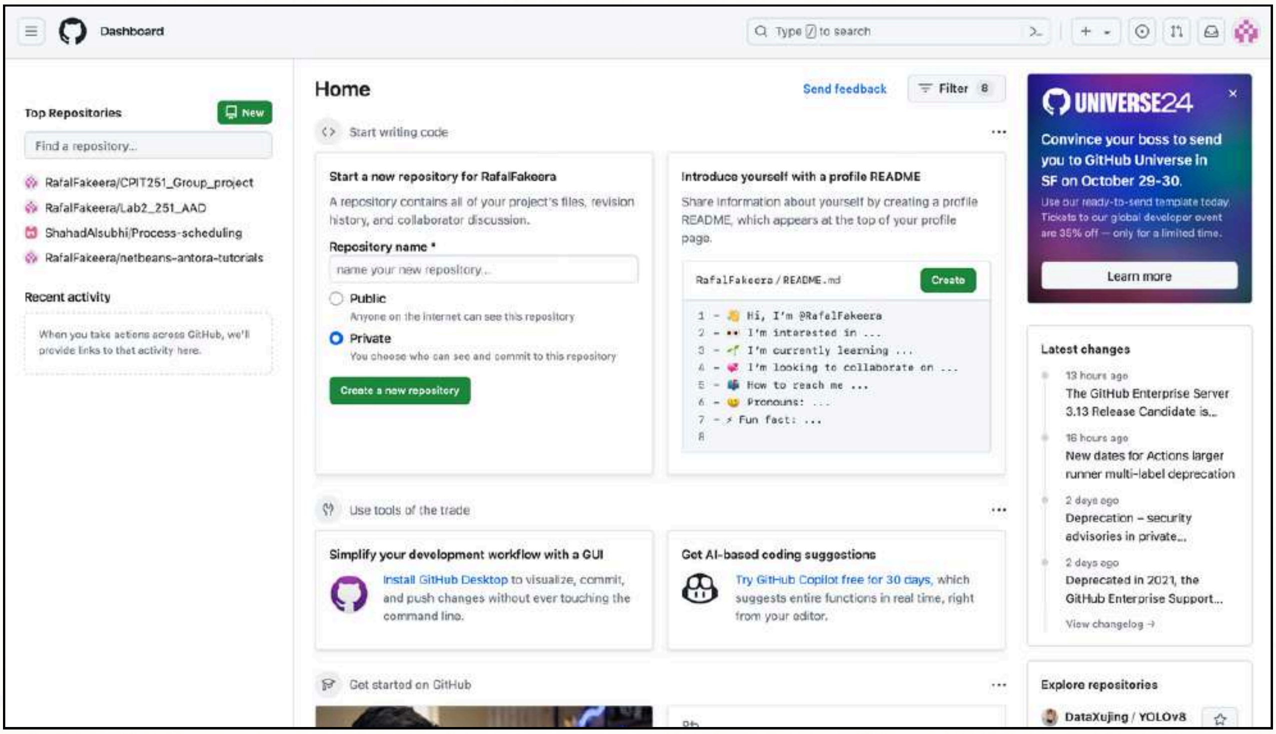
Github

Homepage of GitHub account for each student:

Lojain Alghamdi

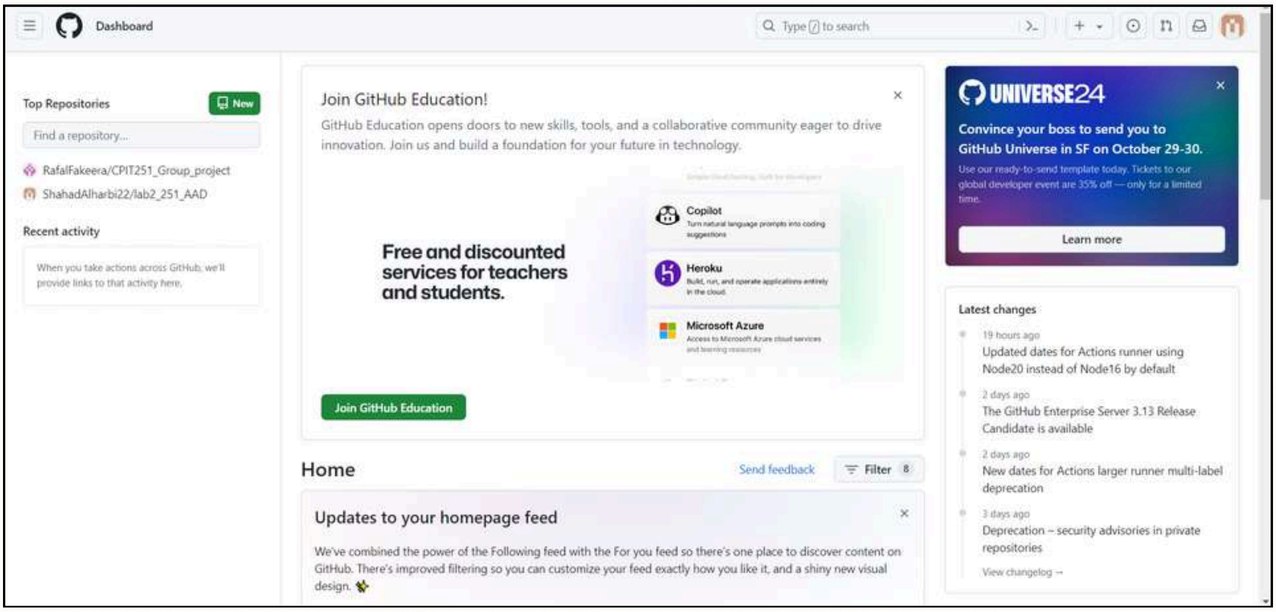


Rafal Fakeera

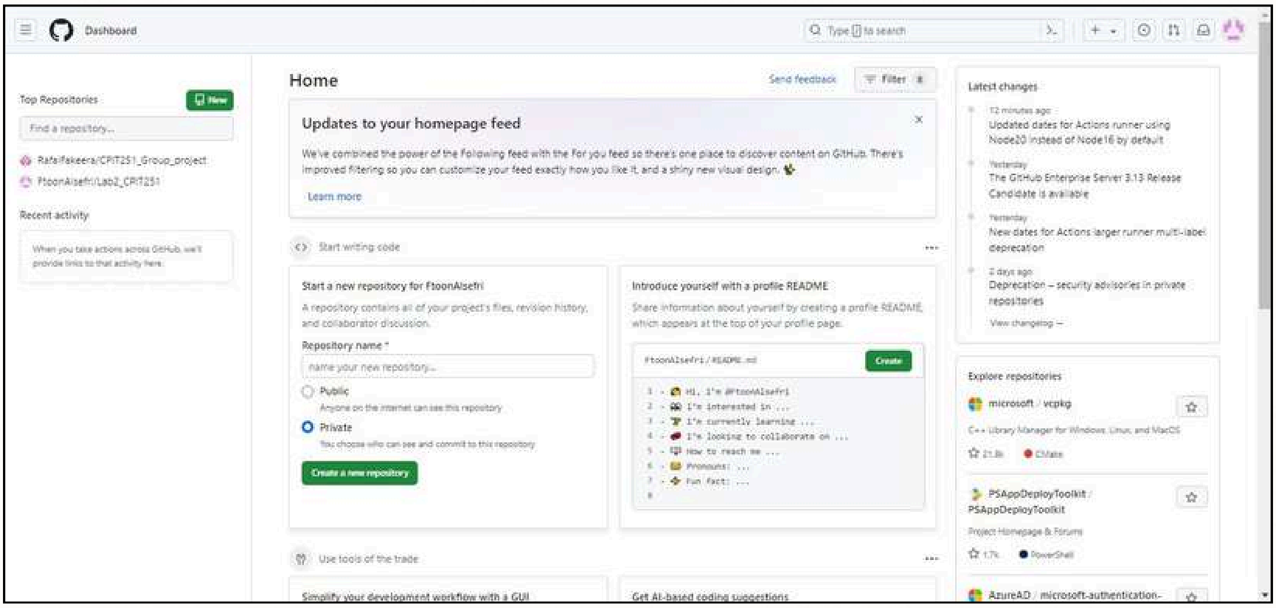


Github

Shahad Alharbi

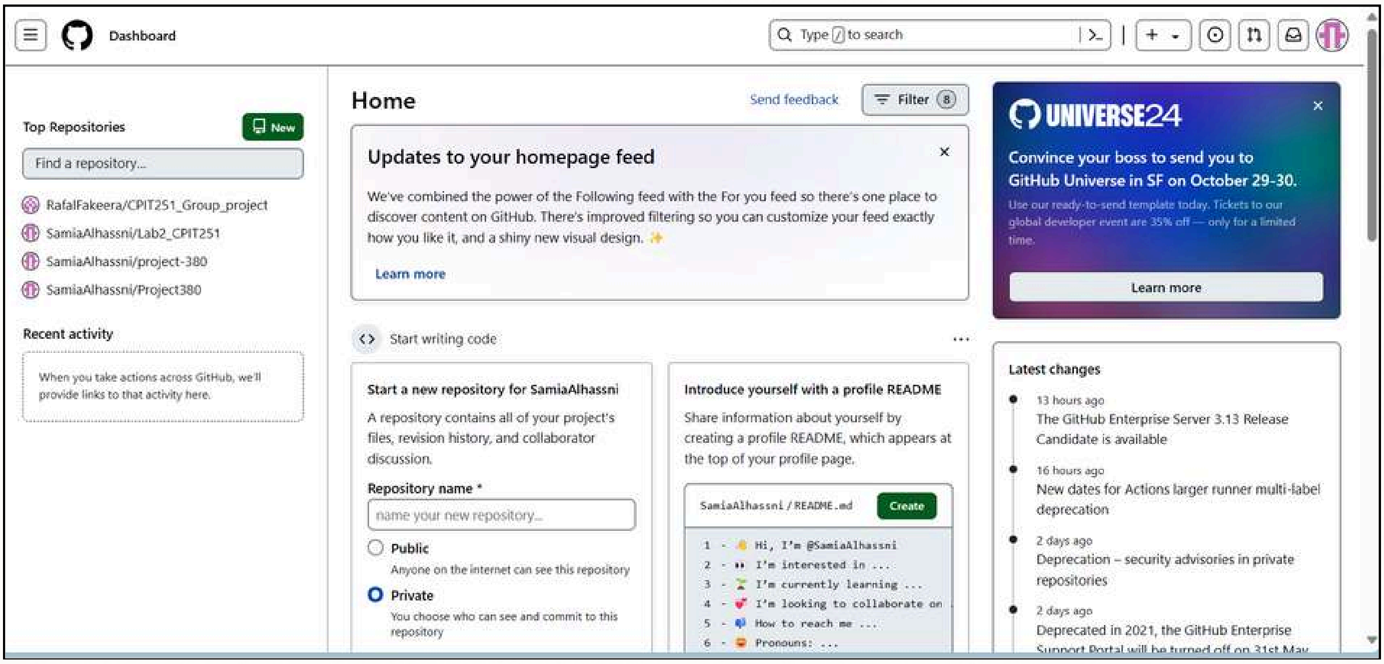


Ftoon Alsefri

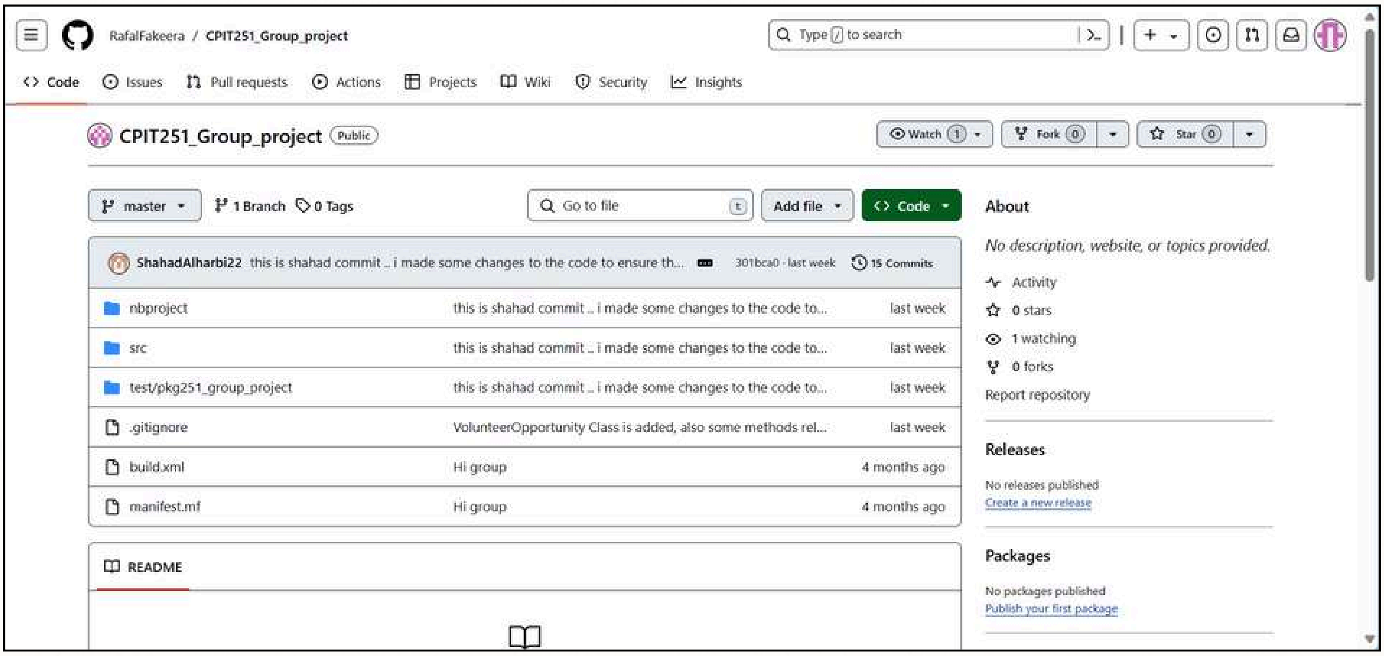


Github

Samia Alhassni

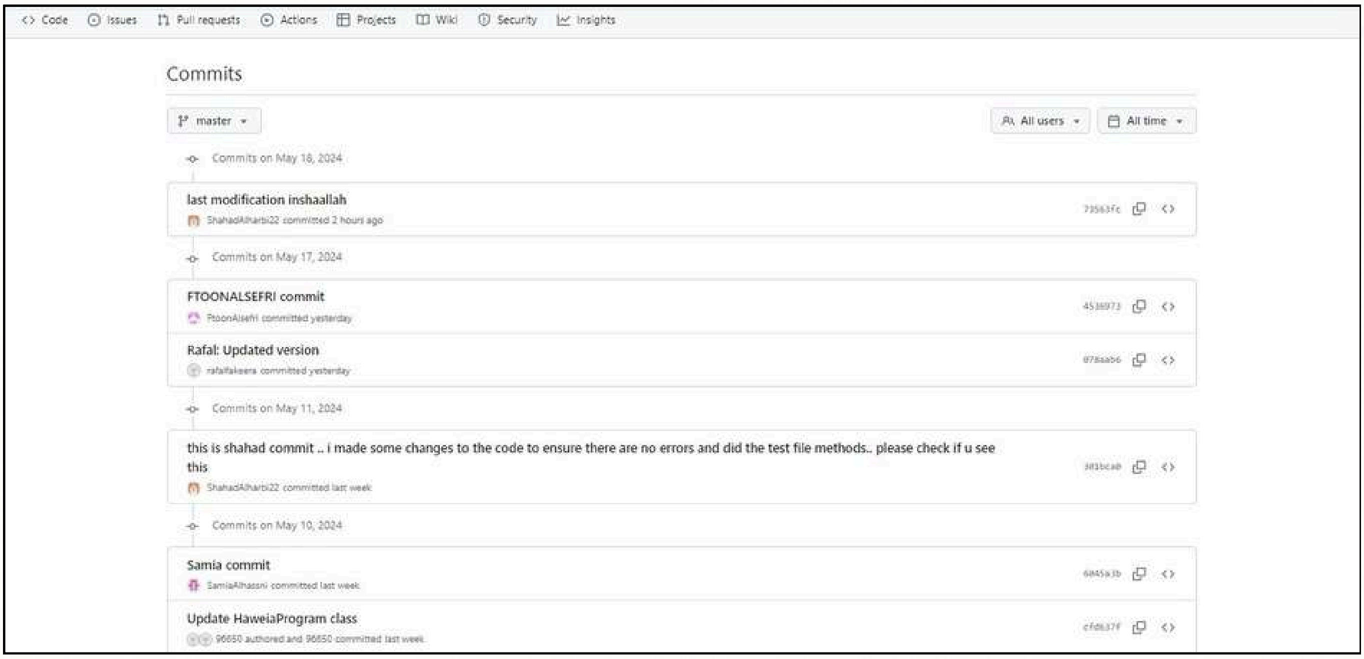


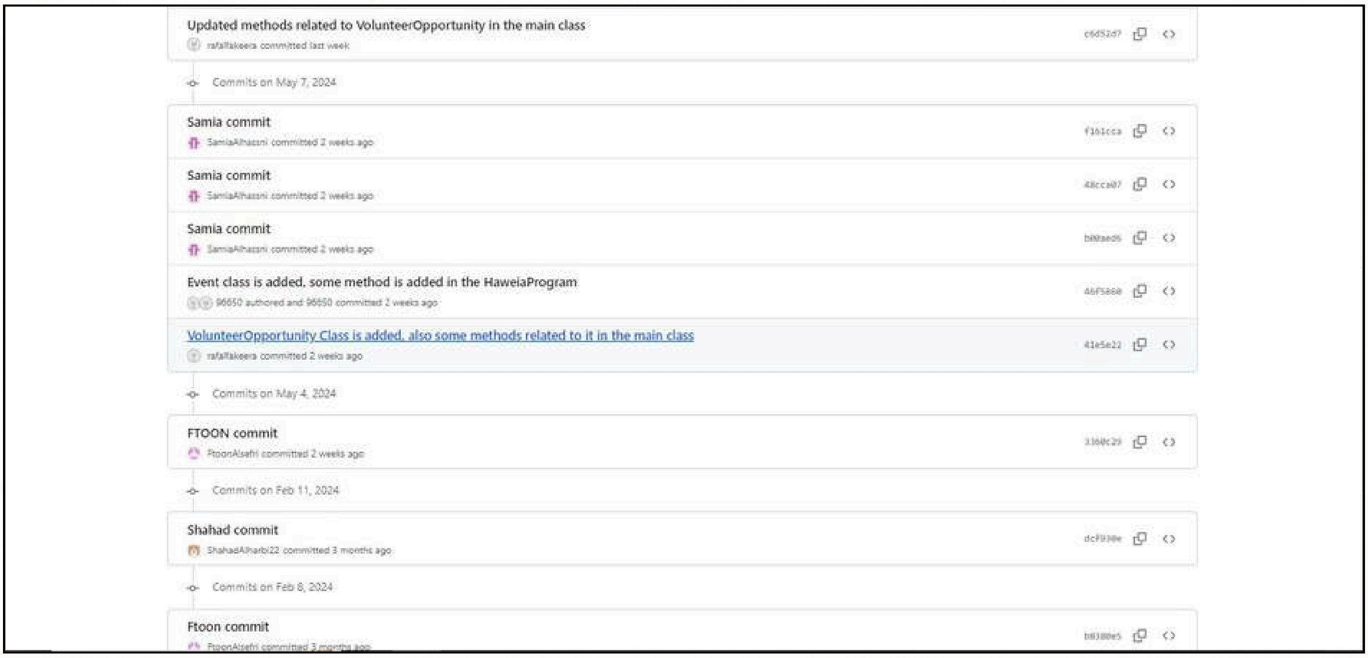
Homepage of the repository of the group:



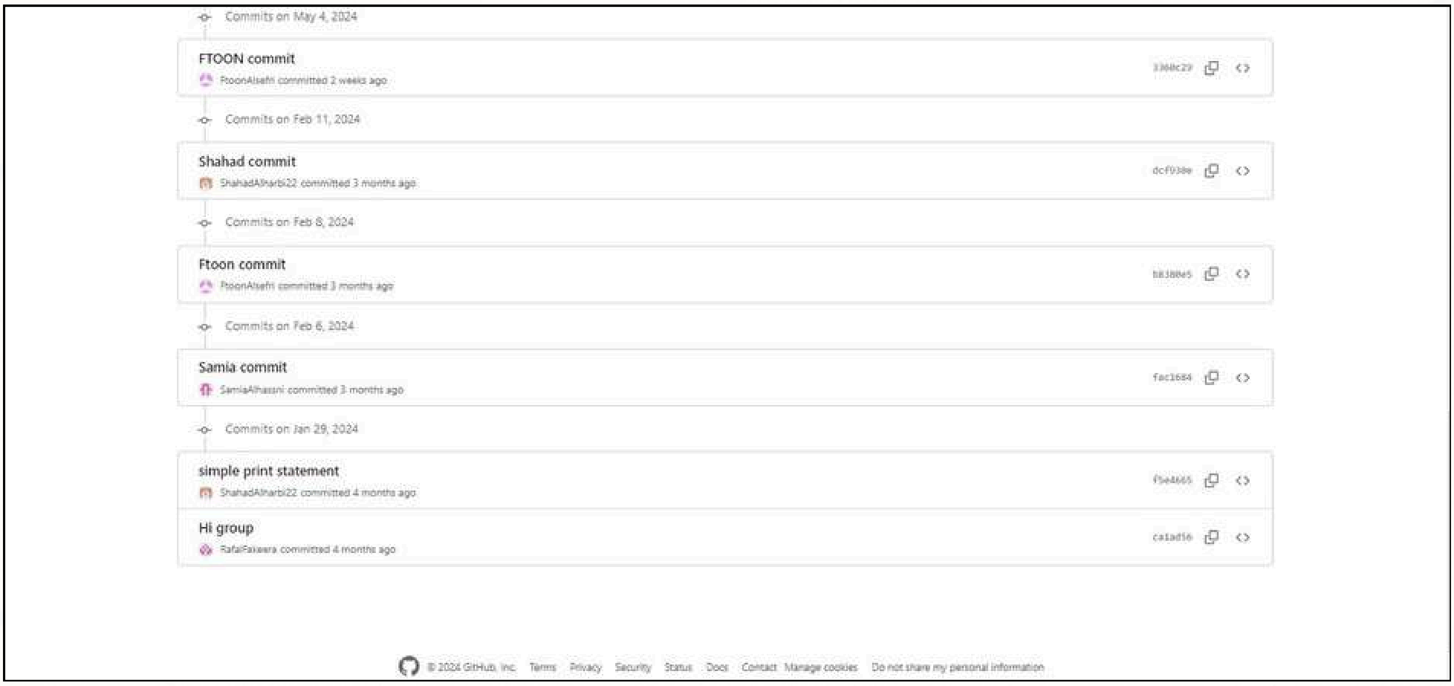
Github

Clear chart for task distribution:





Github


Commits on May 4, 2024
FTOON commit
FtoonAlsefri committed 2 weeks ago
Commits on Feb 11, 2024
Shahad commit
ShahadAlharbi22 committed 3 months ago
Commits on Feb 8, 2024
Ftoon commit
FtoonAlsefri committed 3 months ago
Commits on Feb 6, 2024
Samia commit
SamiaAlhassni committed 3 months ago
Commits on Jan 29, 2024
simple print statement
ShahadAlharbi22 committed 4 months ago
Hi group
RafalFakeera committed 4 months ago
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Future Work

Looking ahead, there are several avenues for future work to further enhance the project's impact and reach. Firstly, incorporating user feedback through iterative development cycles will help refine the user experience and address evolving needs. Also, we can explore providing authentic and immersive experiences for tourists visiting old houses and farms owned by citizens. Rather than relying on virtual technology, tourists will have the opportunity to physically visit these locations and engage in hands-on experiences. For old houses, visitors can explore the historic architecture, interact with period furnishings, and gain insights into the lifestyle of past generations. Similarly, for farms owned by citizens, tourists can participate in agricultural activities, such as harvesting crops, tending to animals, or learning about traditional farming techniques from local farmers. By offering real-world experiences, we can provide tourists with an authentic glimpse into the cultural heritage and rural lifestyle of Saudi Arabia, fostering deeper connections and appreciation for the local community. Lastly, we want to offer a comprehensive travel plans guiding tourists through Saudi Arabia’s popular sites by the utilization of artificial intelligence.

Conclusion

In conclusion, our project has laid the groundwork for promoting cultural tourism in the Kingdom of Saudi Arabia by consolidating information and offering immersive experiences for tourists. Our system “Haweia” was developed using the Waterfall methodology and through the utilization of interviews and questionnaires, we gained valuable insights into stakeholders' perspectives and requirements. The development of Use Case diagrams, Sequence diagrams, and Class diagrams has provided a comprehensive understanding of system functionality and structure. Additionally, the exploration of providing authentic and immersive experiences for tourists visiting old houses and farms owned by citizens presents an exciting opportunity to showcase the rich heritage and lifestyle of Saudi Arabia. Moving forward, our project aims to continue enhancing cultural tourism experiences by leveraging augmented reality (AR) technology to offer hands-on experiences for tourists. By offering authentic glimpses into the Kingdom's cultural heritage and rural lifestyle, we can foster deeper connections and appreciation for the local community, further promoting cultural exchange and understanding.

Appendix

- The user named 96650 in Github is Lojain Alghamdi
- Github link
- Google drive